Powerful Subordinates: Internal Governance and Stock Market Liquidity

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Abstract

In this paper, we examine whether internal governance, the process through which subordinate managers effectively monitor the CEO, can improve a firm's liquidity. Using a measure of internal governance based on the difference in horizons between a CEO and his subordinates, we show that firms with better internal governance have lower information asymmetry and higher liquidity. Further, we show that internal governance is effective in enhancing liquidity for firms with CEOs close to retirement, firms that require higher firm-specific skills, and firms with experienced subordinate managers. Our results are robust to inclusion of conventional governance measures, alternative model specifications, and different measures of internal governance and liquidity.

Keywords: Internal governance, Corporate governance, Subordinate managers, Liquidity.

JEL Classification: G30, G34, G39

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1. Introduction

In this paper we examine whether internal governance affects a firm's liquidity. Internal governance is defined as the process through which subordinate managers can effectively monitor the CEO of the firm. It is the bottom-up governance mechanism that hinges upon subordinate executives' incentive and ability to provide checks and balances to affect corporate decisions. Recent literature shows that subordinates can set up a counter power to self-serving actions of the CEO (Landier, Sraer, and Thesmar 2009; Acharya, Myers, and Rajan 2011; Landier, Sauvagnat, Sraer, and Thesmar 2013; Cheng, Lee, and Shevlin, 2014).

Acharya, Myers, and Rajan (2011) argue that CEOs typically have short horizon and are self-interested, who want to extract the maximum possible rent at the expense of the shareholders and other stakeholders inside the firm. However, there are several reasons to believe that subordinate managers have strong incentives to focus on long-term actions than the CEO. First, subordinate managers have the desire to become the CEO in the future, especially if they are younger and see a sufficient scope for career advancement within the firm.[1] Because of their power to withdraw their contributions to the firm, they can force the CEO to act in more public spirited and far sighted manner (Acharya, Myers, and Rajan 2011). Hence, they have the incentive and the ability to monitor the CEO. Second, subordinate managers have more to lose relative to the CEO. They have more remaining years of employment, which increases the risk of finding a comparable paying job and the potential loss of income (Cheng, Lee and Shevlin, 2014). Further, subordinate managers' opportunity in finding outside position depends on other managers' actions, including the CEO (Fama, 1980).

[1] Acharya, Myers and Rajan (2011) find that about 80% of new CEOs are internally promoted.

In contrast to most empirical research that treats CEO and non-CEO executives as a homogenous group (John and John, 1993; Aggarwal and Samwick, 2003), internal governance literature focuses on how differential preferences between non-CEO top executives and the CEO can affect corporate decisions. Although, previous studies echo the impact of internal governance on investment, firm performance and profitability (Aggarwal, Fu and Pan, 2013; Landier, Sauvagnat, Sraer, and Thesmar, 2013), in this paper, we direct our focus to the impact of internal governance on the information production process and on the firm's stock market liquidity. [2]

Theoretical and empirical evidence reveals that subordinate managers' monitoring is associated with improving firm's information production and decreasing information asymmetry. Fama and Jensen (1983) argue that, due to their expertise, internal managers are the most important source of private and firm specific information and possess a great influence on firm's decision-making process. Landier, Sraer, and Thesmar (2009) also theorize that differential preferences between the CEO and his subordinates can enhance internal monitoring mechanism and foster the information quality. Further, recent studies recognize that the power of subordinate managers come not only from their capacity to withdraw their contributions to the firm, but also from leveraging on their knowledge to push the CEO and the firm to provide more transparent information inside and outside the firm. Cheng, Lee and Shevlin (2014) support this conjecture showing that subordinates can leverage on their knowledge to push the CEO to provide more transparent information about the firm. They empirically show that internal governance, measured by decision horizon of key subordinates and the influence they have on the CEO, increases the financial reporting quality of the firm. Specifically, they find that the extent of real

[2] Aggarwal Fu and Pan (2013) empirically document that internal governance mitigates the under-investment problem and thereby improves firm value. In addition, Landier, Sauvagnat, Sraer, and Thesmar (2013) find that firms with more independent executives from the CEO exhibit a higher level of profitability and higher shareholder returns following large acquisitions.

earnings management decreases as subordinates' horizon and influence increases. The impact of internal governance is also visible in improving the information process inside the firm. For example, Masulis and Mobbs (2011) show empirically that the existence of non-CEO executives on the board improves internal monitoring because of non-CEO executives' firm specific knowledge.

Better information quality and lower information asymmetry is highly correlated with improving liquidity of a firm's shares (Verrecchia 2001; Bardos 2011). Liquidity is negatively related to the level of information asymmetry that results from some traders having informational advantage over other (Glosten and Milgrom 1985; Kyle 1985). Prior literature has shown that governance provisions and external governance can improve liquidity and informativeness of the shares. Chung, Elder and Kim, (2010) show that firms can improve stock market liquidity by adopting corporate governance standards that mitigate informational asymmetries between firm's insiders and external shareholders.[3] In their study, the governance index that is relevant to stock's liquidity includes governance standards mainly related to, board, audit and corporate charter that help improve firm's financial and operational transparency. In addition, institutional investors can exert governance through trading, which increases both stock's liquidity and price informativeness (Gallagher, Gardner and Swan 2013).[4]

However, little is known about whether the differential in horizons within the top management team might affect liquidity of the firm. We hypothesize that, subordinate managers,

[3] Chung, Elder, and Kim (2010) suggest that improving financial and operational transparency decreases information asymmetries between insiders and outside investors as well as among outside investors. Poor transparency insulates and impedes the ability of traders to discern the extent to which management can expropriate firm value through shirking, empire building, risk aversion, and prerequisites (Gompers, Ishii, and Metrick 2003; Bebchuk, Cohen, and Ferrell 2009). Further, providing reliable and accurate information facilitates resource allocation decision and enforcement of contracts for investors.

[4] Shareholders can vote with their feet through trading, even if they face barriers to voice (Admati and Pfleiderer, 2009; Edmans, 2009; Edmans and Manso 2010). Using trading as a governance mechanism is desirable because it improves the value of the firm and leads to a more liquid trading (Edmans, 2009).

the group of the highest-ranked executives in an organization who are responsible for the daily operation of the company, can exert effective monitoring of a self-interested CEO through their contribution to improve a firm's financial and operational efficiency and thereby improving stock market liquidity. This, of course, does not clear off the possibility that the subordinate managers may hold the same short term horizon as the CEO, or that they are unable to pressure the CEO to act in more public spirited way. Thus, the role of subordinate managers on a firm's liquidity through effective monitoring of the CEO is an interesting empirical question.

The previous discussion implies that internal governance depends on the difference in horizons between the CEO and his subordinate managers. Yet, measuring the level of internal governance empirically is quite challenging. The essence of internal governance in Acharya, Myers and Rajan (2011) is linked to the difference in appropriation horizons between the CEO and his subordinates.[5] We follow their approach and use the mean relative age difference between the CEO and the top subordinate managers' as a proxy of the divergence in their horizons within the firm. Recognizing the inherent limitation of a broadly-defined, multi-dimensional measure of horizons such as relative age difference, we have made many adjustments to the raw age difference to account for factors other than appropriation horizons that may be captured in our measure.

Further, although we use relative age difference as the main measure of internal governance, we also use different alternative measures internal governance to corroborate the inference from the main analysis. Motivated by recent literature which relies on executive options grants to capture a CEO's horizon in the firm (Edmans, Fang and Lewellen, 2015; Gopalan, Milbourn, Song and Thakor, 2014), we introduce an alternative measure of internal

[5] The model in Acharya, Myers, and Rajan (2011) is constructed based on a single CEO and a single subordinate. Empirical implications from their model are applied in the current study to include one single subordinate (highest paid non-CEO executive) as well as multiple subordinates.

governance based on the unexercised exercisable options of subordinates relative to that of CEO (options ratio).[6] The options ratio allows us to compute the difference in horizons between a CEO and his subordinates. In addition, we use Bebchuk, Cremers, and Peyer's (2011) CEO pay slice as another alternative measure of internal governance. The CEO pay slice (CPS) or the fraction of CEO compensation to total executives' pay reflects the relative power/contribution of the CEO as well as the extent to which the CEO is able to extract rents. If a CEO dominates power/contribution to the firm, he has no desire to limit his rent extraction or to provide incentives for young managers. In this case internal governance will not be effective as it will be difficult for the subordinate managers to affect the CEO's decisions. Moreover, to account for the difference in team composition among different industries, we use an industry-adjusted age difference as an alternative measure of internal governance.

Using various liquidity measures such as turnover, percentage spread, and Gibbs estimate, we find a positive relationship between internal governance and liquidity. Our results show that subordinate managers, with longer horizon than the firm's CEO, are effective in monitoring the CEO and help increase a firm's liquidity. We show that our results are robust to controlling for CEO and firm characteristics that might affect liquidity of the firm. We show that the positive relation between internal governance and liquidity holds even after controlling for other governance mechanisms studied in prior literature, including analysts following, institutional ownership, and Chung, Elder, and Kim (2010) governance index.[7]

We also conduct additional analyses to further understand the relationship between internal governance and liquidity and ensure the robustness of our results. We find that the

[6] Edmans et al. (2015) and Gopalan et al. (2014) use Equilar dataset that includes the vesting schedule for executives of Russell 3000 firms. We do not have access to Equilar dataset.
[7] See Chung and Zhang 2011; Faure-Grimaud and Gromb, 2004; Chung, Elder, and Kim, 2010; Van Ness, Van Ness, and Warr, 2001.

positive relationship between internal governance and liquidity is non-monotone and it can only be effective under certain circumstances (Landier, Sraer and Thesmar, 2009; Acharya, Myers and Rajan, 2011). We document that internal governance is only effective in improving the liquidity for the firms with CEOs approaching retirement. However, we fail to find similar results for firms with long horizon CEOs. These results are consistent with Acharya, Myers, and Rajan's (2011) conjecture that subordinate managers are effective in monitoring myopic CEOs close to retirement. In addition, we show that internal governance is effective in human capital intensive industries that require subordinate managers to engage in industry and firm specific learning effort prior to their appointment for an executive position. Such firm and industry specific knowledge increases the importance and the power of subordinate managers and also increases their influence on the CEO behavior. We find that only experienced managers are able to exert effective internal governance to increase stock market liquidity of the firm. These results are aligned with the implications of Acharya, Myers and Rajan (2011) model, that is, internal governance is expected to be effective for firms that rely on intensive human capital knowledge.

As previously discussed, the positive association between internal governance and liquidity can arise from the reduction in information asymmetry. Thus, we test directly whether the reduction in information asymmetry serves as a channel through which internal governance increases liquidity. Using Amihud price impact as a proxy of information content (Brennan, Huh and Subrahmanyam, 2013; Amihud, 2002) and Sobel's (1982) mediation test, we show that internal governance significantly increases the informativeness of stock prices. The results are consistent with the argument that subordinates are an important source of firm specific information within the firm (Fama and Jensen, 1983; Core, Holthausen and Larcker, 1999; Masulis and Mobbs, 2011). The ability of subordinates to help the firm provide more reliable information to the market can reduce the information asymmetry and hence improve the liquidity for the firm.

Further, we take a number of steps to mitigate various econometric concerns. One major concern is that internal governance is endogenously determined due to reverse causality or omitted variables. To address concerns that omitted variables drive both past liquidity and internal governance, we include a long list of controls and fixed effect. For instance, including governance index which captures the strength of the board as a control alleviates the concern that a strong board is positively correlated with both liquidity and internal governance. Further, our subsamples results also mitigate this concern because it is arguably harder for an omitted variable or endogenous variable to explain results in both the subsamples and the entire sample. In addition, following Aggarwal, Erel, Ferreira and Matos (2011) and Chung, Elder and Kim (2010), we run regressions with changes in variables. While correlations in changes do not imply causality either, a failure to find correlation in changes is likely to indicate no causal relation. Our results show that changes in internal governance is significantly positively related to changes in liquidity. When we run the changes regression in the reverse direction, using the change in liquidity as the explanatory variable and the change in internal governance as the dependent, the results are statistically insignificant. Overall, our results are not consistent with alternative stories which emphasize the reverse causality from liquidity to internal governance.

To further strengthen our inference on the relationship between internal governance and stock market liquidity, we create a dummy variable to detect changes in top management team that increases the age difference between a CEO and his subordinate managers. We find results consistent with our previous finding of a positive relation between internal governance and liquidity. These steps, taken together, help to establish a more credible positive relationship between internal governance and stock market liquidity.

Our study offers a new way of thinking about the governance role of non-CEO executives within a firm. It differs from prior studies that emphasize the monitoring role of the board, blockholders, analysts, and other external forces. Monitoring is costly and hence, an organizational structure that promotes internal governance through more powerful subordinates could be beneficial for the investors and will improve the efficiency of the capital markets. Internal governance through subordinates, by all means, is not the only form of corporate governance that may affect firm value and liquidity. However, if internal governance is highly effective, there may be less need to rely on other forms of governance (Guo, Lach and Mobbs, 2015).

2. Hypothesis development

In the absence of tight monitoring, CEOs may take actions that are detrimental to the shareholders (Masulis and Mobbs, 2011). In contrast to most empirical research that treats the top management team as one homogenous group, this paper focuses on the role of non-CEO executives in creating a bottom-up internal governance mechanism.[8] In particular, we focus on the impact of internal governance, induced by the difference in horizons between the CEO and his immediate subordinates, on stock market liquidity.

2.1. Internal Governance

Academic literature uses the term "internal governance" to describe different governance mechanisms such as board independence, audit committee independence, shareholders' activism

[8] Graham, Harvey, and Puri (2013) suggest that managers, other than the CEO, perform important functions, especially in large and complex firms.

and institutional holding.[9] However, Acharya, Myers, and Rajan (2011) and Landier, Sraer and Thesmar (2009) depart from these conventional views and use the term internal governance to describe the bottom-up governance mechanism asserted from the CEO's immediate subordinates. According to Acharya, Myers, and Rajan, (2011), the firm is a composition of diverse agents with different horizons, interests, and opportunities for misappropriation and growth. They suggest that the CEO is not always a faithful agent of shareholders and other stakeholders of the firm. Although the CEO is the top ranked executive within a firm's managerial hierarchy, he is not the sole productive asset in the firm. The CEO needs the cooperation of his subordinates in order to operate the firm. An older CEO with a shorter employment horizon than his subordinates may extract rents at the expense of the shareholders and other stakeholders inside the firm. However, younger subordinate managers, in turn, may have a long-term interest in the firm's prospects, especially if they see a sufficient scope for career advancement within the firm (i.e. become the next CEO) (Prendergast, 1999). If subordinate managers see that the CEO will leave nothing behind, they are reluctant to exert effort and the firm's cash flow will fall significantly.

Landier, Sraer, and Thesmar (2009) suggest that internal governance can forster the use of objective information and improves the information production process inside the firm. Fama and Jensen (1983) argue that, internal managers are the most important source of private and firm specific information which increase their power and ability to affect on firm's decision. More recently, Cheng, Lee and Shevlin (2014) provide direct evidence about the importance of subordinate managers in improving firm's reporting quality. They show that subordinate managers horizon is negatively related to earning management practices of the firm. Also,

[9] See Johnstone, Li, and Rupley (2011), Cremers and Nair (2005), and Chung, Elder and Kim (2010).

Masulis and Mobbs (2011) show empirically that the existence of non-CEO executives on the board improves internal monitoring because of non-CEO executives' firm specific knowledge.

The constraints, which the two parties inside the firm's top executive team, CEO and his subordinate managers, impose on each other ensure that the firm can function and survive, even if outside governance is weak. As the previous discussion implies, the importance of subordinate managers is not limited to contribution withdrawal, but also associated with their firm specific knowledge to elicit better decision-making from the CEO, particularly, to produce and disclose more reliable and accurate information about the prospects of the firm. Thus, this bottom up internal governance mechanism may increase the financial and operational transparency in the financial market and therefore enhances liquidity. [10]

The following hypothesis tests the relationship between internal governance exercised by non-CEO executives and stock market liquidity.

H1: There is a positive relationship between internal governance and stock market liquidity.

2.2. CEO Horizon and Internal Governance

Gibbons and Murphy (1992) and Xu (2011) document that a CEO with a shorter remaining horizon within the firm loses incentives to focus on the long term firm performance. Such myopic CEOs pass investments in projects with positive long term payoffs in order to boost contemporaneous earnings (Stein, 1989), overinvest to signal that they have investment

[10] The subordinate managers can use their knowledge to limit CEO's rent seeking behavior and to provide more information to the market participants. Hence, they can reduce the agency cost between insiders and outsiders and also improve the corporate governance. Previous literature provides evidence that subordinate managers can enhance the operational and financial transparency of the firm and effectively constrain value destroying behavior on part of the CEO. For example in their theoretical model Fama and Jensen (1983) suggest that one of the most important sources of firm specific information to board is the information that comes from non-CEO executives. In addition, previous literature shows empirically that non-CEO executives improve internal monitoring because of their firm specific information (Core, Holthausen and Larcker, 1999 and Masulis and Mobbs, 2011). We take a step further to investigate whether subordinate managers can use their power and knowledge to ensure that the CEO and the firm provide more transparent information to the market, which in turn reduces the information asymmetry and improves the stock market liquidity.

opportunities (Bebchuk and Stole, 1993), or take excessive risk (Chevalier and Ellison, 1999). Aforesaid CEO's myopic behavior is also visible in practice. One example is the Merck scandal. During the last quarter of 2004, Merck had to pull Vioxx off the market due to the concern that links Vioxx to increased risk for heart attack or stroke, resulting in 28% decline in the stock price. Despite the firm's poor stock performance in 2004, Ray Gilmartin, the 64-Year-Old CEO, received not only his base salary but performance based bonuses worth $37.8 million. Ray Gilmartin was scheduled to retire at the age of 65, ended up leaving his position 8 months prior to his mandatory retirement age.

Acharya, Myers, and Rajan (2011) suggest that the internal governance is only effective for the firms with CEOs close to retirement.[11] Hence, we test the following hypothesis:

H2: Internal governance is only effective in improving liquidity for firms with shorter horizon CEOs.

2.3. Firm-specific Knowledge and Internal Governance

Acharya, Myers, and Rajan (2011) conjecture that subordinate managers can be a reliable part of internal governance only if they have a stake in the future of the firm. This typically requires that firm-specific knowledge and experiences to be highly valued and/or a high cost of exiting the firm. Thus, those non-CEO executives with firm and industry specific knowledge are likely to have a vested interest in the firm and are able to exert greater influence on the CEO (Landier, Sauvgnat, Sraer and Thesmar, 2013) and may effectively enhance firm's transparency.

In human capital intensive industries, managers are required to engage in industry and firm-specific learning efforts prior to their appointment for an executive position, which increases their importance, power, and influence on the CEO. We conduct a test to examine the

[11] Acharya, Myers, and Rajan (2011) also report that executives other than the CEO are, on average, four years younger than the CEO

impact of firm specific/industry specific learning on the effectiveness of internal governance. Pantzalis and Park (2009) provide a rank of Fama and French 48 industries based on excess value of human capital.[12] For example, the pharmaceutical industry, a research-intensive and highly innovative industry that depends on intellectual capital as a source of growth, scores the highest rank. Other top-ranked industries, such as, natural gas, real estate, financial services, precious metals, communication and energy, share similar characteristics. On the other extreme, the capital intensive industries, such as manufacturing industries, that are less innovative and do not depend heavily on intellectual capital, are ranked in the bottom tier. We follow Pantzalis and Park (2009) to rank firms' human capital intensity and test the following hypothesis:

H3: Internal governance is only effective in improving the stock market liquidity for firms that require firm specific skills and knowledge.

2.4. Subordinate Managers' Experience and Internal Governance

Given the pivotal role played by non-CEO executives, understanding different executives' traits lead to a more accurate assessment of the effectiveness of internal governance in enhancing the firm transparency and stock market liquidity. Acharya, Myers, and Rajan (2011) suggest that experience of subordinate managers' is very critical not only for the CEO promotion but also for effective monitoring. Consider a firm with a two-level managerial hierarchy: at the top of this hierarchy is a CEO who is approaching retirement, in the second layer is a younger subordinate manager who will become the CEO next period. If the subordinate manager lacks the needed experience he might not be considered for CEO promotion and hence he will not be motivated

[12] Pantzalis and Park (2009) measure excess value of human capital as follows: first, they compute the industry-median value for the ratio of market value of common equity to total number of employees (EV). Then, they multiply the industry median EV by the firm's number of employees to obtain an imputed market value of human capital.

enough to monitor the CEO actions. To test the impact of subordinate managers' experience on the efficacy of internal governance, we test the following hypothesis.

H4: Internal governance is only effective in improving the stock market liquidity for firms with experienced subordinate managers.

3. Data and Methodology

Our sample includes 7 years of data for S&P 1500 firms for the period from 2001 to 2007. Standard & Poor's ExecuComp database provides annual data on CEO and subordinate managers' ages, appointment dates, dates for leaving the firm, compensation, and other attributes. Following Acharya, Myers, and Rajan (2011) we limit our sample to include the top 5 subordinate managers including the CEO.[13] Stock prices, closing bid and ask prices, trading volume, and shares outstanding are derived from the Center for Research in Stock Prices (CRSP) database. Data on analyst coverage is extracted from the Institutional Brokers' Estimate System (I/B/E/S) dataset. Data on institutional holdings is collected from the 13F fillings summarized in the CDA/Spectrum database. Using Institutional Shareholder Service (ISS) database, we extend Chung, Elder, and Kim (2010) Gov-Index beyond their sample period of 2001 to 2004, to cover the period from 2001 to 2007. Gov-Index includes 24 governance standards across six categories that are most closely related to financial and operational transparency and investor protection.[14] Finally, we obtain financial and accounting data such as total assets, intangible assets, dividend, and R&D expenditure from Compustat database.

[13] Acharya Myers and Rajan (2011) show in Table II, that nearly 80% of new CEOs are appointed from the top four executives in the firm in the previous year. Some firms in our sample have less than four subordinate managers. In an alternate measure we estimate internal governance by including only one single subordinate manager who receives the highest pay among all subordinate managers in the firm.

[14] Similar to Brown and Caylor (2006), Chung, Elder and Kim (2010) create their index (Gov-Index) for each firm by awarding one point for each governance standard that is met. Gov-Index includes governance standards related to audit committee, Board independence, Charter and firm's Bylaw, executive compensation, ownership constructions and state law.

The trading data includes some trades with zero trading prices and bid and ask quotes for the locked and crossed markets (bid price less than/equal to ask price). These observations are removed from the final sample. We also apply the following data filters, which are standard in the market microstructure literature (Huang and Stoll, 1996), to clean the data entry errors and outliers. We delete 1) quotes if either bid/ask is negative, 2) if quote changes by more than 10% compared to the last quote, 3) stocks with average annual price less than $5 and greater than $500, and 4) stocks that are not included in S&P 1500 index, Compustat, Center for Research in Security Prices (CRSP), CDA/Spectrum, I/B/E/S, or the ISS database. The variables used in this study and their sources are summarized in Appendix A.

3.1 Internal Governance Measures

3.1.1 Difference in Horizons

Crucial to the internal governance in Acharya, Myers, and Rajan (2011) model is the difference in appropriation horizons between the CEO and his subordinates. Prior literature commonly uses age as a proxy for employment horizon (Brickley et al., 1999, Gibbons and Murphy 1992, Dechow and Sloan 1991, Matejka et al., 2009). Although, tenure in the firm is an alternative way to proxy horizon but it has three major problems. First, it reflects only the past horizon and may not infer anything about the executives' expected future horizon. Second, it ignores managers' experiences outside the current firm. Third, it ignores the cumulative learning and experiences of managers beyond their executive position. Age, as a proxy of the horizon within the firm, can circumvent these concerns. Hence, we use the average age difference between the CEO and other top four executives as a proxy for the difference in horizons within the firm and thus, the level of internal governance. Our first internal governance measure is calculated as follows.

$$Internal\ Governance\ (Age\ difference) = CEO's\ Age_{i.t} - Subordinate\ Managers'\ Age_{i,t} \quad (1)$$

where $CEO's\ Age_{i.t}$is the age of CEO and $Subordinate\ Managers'\ Age_{i,t}$ is the mean age of the top four subordinate managers for firm *i* at year *t*. We follow Acharya, Myers and Rajan (2011) and Bebchuk, Cremers and Peyer (2011) by including only the top 4 non-CEO executives to calculate our internal governance measure. This also allows us to avoid loss of data points due to missing observations on ages of executives beyond the top four. The use of age difference instead of the raw ages, rules out the possibility that this measure may proxy other attributes of CEO or subordinates such as; sophistication, risk taking, education, or experience.[15]

3.1.2 Industry Adjusted Difference in Horizons

The composition of management teams depends on the nature of the industry. For example, industries in their infancy, such as computer software industry, would have companies with younger management teams and younger CEOs as compared to mature industries. Adjusting the internal governance measure to account for industry differences might add additional important insights to our study. To account for the differences of team composition among the different industries, we develop an industry-adjusted measure of internal governance. The industry-adjusted measure of internal governance is calculated as follows.

$$Internal\ Governace\ (Industry\ Adjusted\ Age\ difference) =$$
$$Firm\ level\ age\ difference_{i,t} - Industry\ level\ age\ difference_{industry,t} \quad (2)$$

A positive (negative) value indicates that the relative age difference between the CEO and his subordinates' managers for a firm is greater (less) than the industry average.

3.2. Liquidity Measures

[15] Raw age can be directly related to risk aversion, education, or experience but the age difference between CEO and subordinates controls for the individual attributes (see, Ang, Cole and Lawson, 2010; Lundstrom, 2002; Wiersema and Bantel, 1992; Bantel and Jackson, 1989).

Kyle (1985) notes that ‘‘liquidity is a slippery and elusive concept, in part because it encompasses a number of transactional properties of markets, these include tightness, depth, and resiliency.’’ Empirical liquidity measures span from direct trading costs (tightness) measured by spreads, to indirect trading costs measured by price impact. The literature provides a variety of measures and proxies to estimate liquidity. We use three different measures of liquidity: turnover, percentage spread, and Gibbs estimate proposed by Hasbrouck (2009). Percentage spread and Gibbs estimate are inverse measures of liquidity while turnover is a direct measure of liquidity.

3.2.1. Turnover

Turnover is the daily volume relative to the number of shares outstanding. We use the firm-year average of this measure. Since volume on NASDAQ is known to be overstated as a result of trades between dealers, we divide volume on NASDAQ-listed firms by 2 to get to an adjusted turnover measure (Atkins and Dyl, 1997 and Lipson and Mortal, 2009).

$$Turnover_{i,t} = \frac{Daily\ Volume_{i,t}}{Shares\ Outstanding_{i,t}} \tag{3}$$

3.2.2. Percentage Spread

Percentage spread is commonly measured as the difference between the best ask quote and the best bid quote relative to the bid-ask midpoint.

$$Percentage\ Spread_{i,t} = \frac{Ask_{i,t} - Bid_{i,t}}{(Ask_{i,t} + Bid_{i,t})/2} \tag{4}$$

where $Ask_{i,t}$ is the closing ask price and $Bid_{i,t}$ is the closing bid price for stock *i* for day *t*.

3.2.3. Gibbs Estimate

Roll (1984) proposes a method to estimate bid-ask spreads from the time series of daily price changes. Specifically, he notes that positive spreads will induce negative serial correlation in transaction price changes and that spreads can be estimated from that serial correlation. One of the limitations of the measure is that estimates of spreads are negative when stock price changes are positively correlated. Hence, we use a modified Gibbs estimate, developed by Hasbrouck (2009) that addresses this econometric problem of the original Roll's measure.[16]

3.3. Control Variables

To control for other governance mechanism, we extend our analysis to include other conventional governance mechanisms that are established in the literature and are known to affect stock market liquidity.[17]

3.3.1. External Governance

The two widely accepted external corporate governance measures are institutional ownership and analyst coverage of the firm. Chung and Zhang (2011) find that institutional investors prefer stocks of better-governed firms for fiduciary responsibilities, monitoring costs and liquidity reasons. A block holder participates in value increasing activities in proportion to his equity stake in a firm. A larger stake increases his benefit from higher firm value (Faure-Grimaud and Gromb, 2004) and alleviates the free-rider problem pervasive in firms with passive dispersed investors (Shleifer and Vishny, 1997). Recent literature suggests that block holders not only add value through direct intervention, but also can improve the firm value through trading (McCahery, Sautner, and Starks, 2010; Admati and Pfleiderer, 2009; Edmans, 2009; Edmans and Manso, 2010).

Analyst coverage has two competing effects on the corporate governance. The first perspective argues that the firms that are widely followed by analysts may be pressured to adopt better corporate governance which results in less information asymmetry and higher liquidity

[16] Gibbs estimate of the Roll's (1984) measure is available at Joel Hasbrouck's website: http://people.stern.nyu.edu/jhasbrou/.

[17] In this paper, we define internal governance as the governance mechanism that works inside the managerial structure of the firm. Under this definition, we assume that other stakeholders outside the managerial structure, such as institutional investors, are external forces.

(Brennan, Jegadeesh, and Swaminathan, 1993; Brennan and Subrahmanyam, 1995; Lang, Lins and Miller, 2004). Alternatively, analysts have incentive to build their own reputation to increase their compensation. Hence, they follow the stocks with greater information asymmetry among market participants or stocks with lower liquidity for which the marginal benefit of information production is the greatest (Chung, Elder, and Kim, 2010; Van Ness, Van Ness, and Warr, 2001; Chung, McInish, Wood, and Wyhowski, 1995).

3.3.2. Corporate Governance Index

Firm's board independence, audit committee independence and other governance provisions can also play an important role in the firm's corporate governance. Gompers, Ishii, and Metrick (GIM) (2003) provide one of the most highly cited governance index. The GIM index was designed primarily to capture anti-takeover provisions in a firm's charter, bylaws, and state law. However, Chung, Elder and Kim (2010) argue that the GIM governance index is not appropriate in understanding the effect of governance on liquidity. They propose a new governance index (Gov-Index) which captures governance standards related to the independence of the audit committee, independence and effective functioning of the board, executive and director compensation and ownership, provisions in the firm's charter and bylaws, and incorporated state provision. Chung, Elder, and Kim (2010) document a positive relationship between Gov-Index and stock market liquidity. In this paper, we use Gov-Index to control for other governance provisions that might affect stock market liquidity.

3.3.3. Other Control Variables

Level of monitoring and market liquidity could be spuriously correlated because they are related to a common set of variables such as the size of the firm. Including the variables that are related to both level of monitoring and market liquidity in a regression model can reduce the possibility of spurious correlation. For example, Chung and Zhang (2011) and Chung, Elder, and

Kim (2010) document that the larger firms may simultaneously exhibit better governance structure due to higher investor interest and lower spreads due to smaller adverse selection risks (e.g., more information is available on larger firms). We include the following control variables: closing price, research and development expenses, firm size, trading volume, return volatility, intangible assets, dividend per share, and return on S&P 500 index. We provide further details on these variables in section 4.2.

4. Empirical Results

4.1. Descriptive Results

Table 1 reports the descriptive statistics for the key variables included in our study. The mean CEO age is 54.68 years. The youngest CEO in our sample is 34 years old, while the oldest is 91 years old. The mean age for the top four subordinate managers is 50.69 years; the youngest is 33 years old while the oldest is 81 years old.[18] The mean age difference between the top 4 subordinate managers' ages and the CEO's age is 4 years. The distribution of age difference ranges from firms with subordinate managers older than the CEO by 25 years, to firms with a CEO older than his subordinate managers by 39.25 years. Our age difference statistics are comparable to those reported by Acharya, Myers and Rajan (2011).[19] Table 1 further reports an average Gov-Index of 12.49, indicating that, on average, the sample firms meet approximately half of the governance standards.[20] In addition, our sample firms have, on average, 71% institutional ownership and 10.87 analysts following their stocks. Table 1 also shows that the

[18] The CEO's age is available for 7694 observations and subordinate managers' age is available for 7556 observations for the period from 2001 to 2007. In Table 1, we include statistics on CEO and subordinate managers only for firms that have at least one liquidity measure.

[19] Acharya, Myers and Rajan (2011) provide the distribution of CEO and Subordinated managers' ages in Table 1 over the period of 1992- 2008. They report a mean CEO age of 55.6 years, a mean subordinated managers' age of 51.6 years, and the mean age difference of 4 years.

[20] Chung, Elder and Kim (2010) reports that the average Gov-Index is 11.49 for their sample firms, the minimum is 3 while the maximum is 20.

means of turnover, percentage spread, and Gibbs estimate are 0.26%, 0.25% and 0.66% respectively. Descriptive statistics for other control variables are also reported in Table 1.

Insert Table 1

4.2. Liquidity and Internal Governance

To test our various hypotheses, we regress turnover, percentage spread, and Gibbs estimate on internal governance and other control variables using annual pooled cross-sectional and time-series data. Previous studies show that a significant portion of cross-sectional and time-series variation in liquidity can be explained by select stock attributes (see McInish and Wood, 1992; Chung, Van Ness, and Van Ness, 1999; and Stoll, 2000). To isolate the effect of internal governance on liquidity, we control for stock's average daily closing price (in logarithm), return volatility, dollar trading volume (in logarithm), and dividend per share. S&P 500 return is included to control for the market conditions in the regression model.

Previous research shows that firm size can be positively correlated with both better governance and lower liquidity (Chung, Elder and Kim, 2010; Chung, and Zhang, 2011). To examine whether corporate governance has an independent, direct impact on liquidity, we control for firm size, measured by the book value of total assets, in the regression model. Further, we control for asset tangibility and R&D expenditure. Asset tangibility could reduce asymmetric information as payoffs on tangible assets' are easier to value. In contrast, high R&D intensity may increase asymmetric information because payoffs from R&D are difficult to predict. Based on these considerations, we estimate the following regression model:

$$\begin{aligned}&\textit{Turnover}_{i,t} \textit{ or Percentage Spread}_{i,t} \textit{ or Gibbs estimate}_{i,t} = \beta_0 + \beta_1 \textit{Internal Governance (Age Difference)}_{i,t} + \beta_2 \textit{Gov-Index}_{i,t} + \beta_3 \textit{Institutional Ownership}_{i,t} + \beta_4 \textit{Analyst Coverage}_{i,t} + \beta_5 \\ &\textit{Log(Closing Price}_{i,t}) + \beta_6 \textit{R\&D Expenditure}_{i,t} + \beta_7 \textit{Total Assets}_{i,t} + \beta_8 \textit{Log(Trading Volume}_{i,t}) + \\ &\beta_9 \textit{Volatility}_{i,t} + \beta_{10} \textit{Intangible Asset}_{i,t} + \beta_{11} \textit{Dividend}_{i,t} + \beta_{12} \textit{Market Return}_{i,t} + \varepsilon_{i,t} \qquad (5)\end{aligned}$$

where *Turnover* is average daily volume relative to shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and mean of top 4 *Subordinate Managers' Ages*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P 500 index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses.

Insert Table 2

Table 2 summarizes our main results from the regression analysis.[21] We use three different model specifications to analyze the relationship between liquidity and internal governance. The first model uses internal governance, measured by the relative age difference, as the sole level of monitoring. The second model adds the external governance as the additional level of monitoring, and the final model accumulates the total effect of all governance measures discussed earlier. The results show that the coefficients on relative age difference are negative and statistically significant for both the percentage spread and Gibbs estimate, and positive and statistically significant for turnover across different model specifications. Hence, firms with

[21] In later sections, we tested the robustness of our findings by using alternate measures of internal governance and find results consistent with the ones presented here.

higher age difference between the top subordinate managers and the CEO (or a higher divergence of appropriation horizons) are more liquid. These results highlight the importance of subordinate managers in internal corporate governance. Subordinate managers have a longer horizon in the firm than the CEO, and thus are motivated to monitor the incumbent CEO. In addition, they are able to leverage on their power and knowledge to help the CEO and the firm to provide more reliable and accurate information to the financial market to keep the firm liquid and attractive to stock market investors. We further test the robustness of our findings by using an alternate measure of internal governance calculated as the age difference between the age of the CEO and the age of the highest paid subordinate manager and results remain qualitatively similar.[22] In sum, these results support Hypothesis 1 and are consistent with Acharya, Myers, and Rajan's (2011) theoretical model.

Consistent with previous studies, we also find that liquidity is significantly and positively related to closing price, asset tangibility, return volatility, and dollar volume (McInish and Wood, 1992; Chung, Van Ness, and Van Ness, 1999; and Stoll, 2000). Further we find that liquidity is negatively related to R&D expenditures, volatility, and dividends. Our results are consistent with the findings of Chung, Elder, and Kim (2010) in the sense that a higher R&D expenditure may increase asymmetric information, because payoffs from R&D expenditures are difficult to predict, resulting in lower liquidity. Our results also support the findings of Banerjee, Gatchev, and Spindt (2007) that owners of more (less) liquid common stocks are less (more) likely to receive dividend. They suggest that cash dividends reduce investor dependence on the liquidity of the market, and hence, liquidity and dividends are negatively related.

Consistent with Chung and Zhang (2011) and Chung, Elder and Kim (2010) we find that institutional ownership is significantly positively related to liquidity. Institutional investors

[22] Results available upon request.

provide effective monitoring and reduce the information asymmetry between insider and liquidity providers, resulting in an increase in liquidity.

Further we find a significant and negative relationship between liquidity and number of analysts following. One possible interpretation is that analysts have incentives to build their own reputation by following firms with a greater information asymmetry. These results are consistent with the findings of Chung, McInish, Wood, and Wyhowski (1995) and Chung, Elder, and Kim (2010).

In our final model we run a pooled regression by including all measures of monitoring in one regression equation, we find that the coefficient for Gov-Index is significantly positive for Turnover and significantly negative for Gibb's C-estimate and percentage spread. These results are in line with the conjecture that better governance leads to a higher stock market liquidity. More importantly, our measure of internal monitoring, the relative age difference between CEO and subordinate managers, is still significant and positively related to liquidity, even after controlling for all other levels of corporate governance.

4.3. Testing the Channel: Internal Governance and Information Content

In this paper, we conjecture that the channel through which internal governance affects liquidity is that better internal governance improves operational and financial transparency and decreases the information asymmetry. Subordinate managers, with the ability to produce more reliable information for the firm to disclose to the market, can actively reduce the information asymmetry and hence improve the liquidity for the firm. We present a direct test of the impact of internal governance on the information content of prices.

Amihud price impact (2002), which is based on Kyle (1985) lambda and the daily price impact, has been used as a measure of information content in stock prices. It measures the daily price response associated with one dollar of trading volume, thus serving as a measure of price impact. Hasbrouck (2009) shows that Amihud's (2002) measure is most highly correlated, 0.82 correlation, with benchmark price impact measures based on intraday data. Price impact, in turn, reflects the information content and trade frictions. Brennan, Huh, and Subrahmanyam (2013) decompose Amihud's (2002) premium. Consistent with Sarkar and Schwartz (2009), they find that Amihud's (2002) premium is associated with the presence of information asymmetry.

Brennan, Huh, and Subrahmanyam (2013), suggest that, since Amihud (2002) treats the dollar volume of trading (the product of firm size and turnover) as a measure of trading activity, the illiquidity premium can be estimated using a measure that relies on turnover as the measure of trading activity and accounts firm size effects separately. As our main model controls for firm size, we follow Brennan, Huh, and Subrahmanyam (2013) and calculate the turnover version of Amihud illiquidity to proxy the level of information asymmetry. We also follow Karoyli, Lee and van Dijk (2012) and add a constant and take the log to reduce the impact of outliers.

$$Turnover\ Version\ of\ Amihud\ Price\ Impact_{i,t} = log\ (1 + \frac{|Ret_{i,t}|}{Turnover_{i,t}}) \qquad (6)$$

To examine whether information content is an important channel between internal governance and firm's liquidity, we use Sobel's (1982) test of mediation. The goal of a mediation analysis is to assess whether information content serves as a significant channel through which internal governance affects liquidity (the dependent variable)[23].

[23] The first step in the mediation analysis is to show that dependent variable of interest, liquidity, is related to the independent variable, internal governance: $Liquidity = \beta_1 + \tau\ Internal\ Governance_i + \varepsilon_1$. Next the mediator and original independent variable are included in the same regression along with the dependent variable of interest (e.g., liquidity). $Liquidity = \beta_2 + \tau^*\ Internal\ Governance + \varphi\ \ Information\ content + \varepsilon_2$. $\beta_1 and \beta_2$ denote intercept for model 1 and, 2 respectively. τ, τ^* and φ denote the relationship between independent and dependent variables and ε_1, and ε_2, denote unexplained variability. If the mediator variable, information content, mediates the relation between the dependent variable, liquidity, and the original independent variable, (internal governance), then the significance of the original independent variable will be reduced over the first stage regression and the mediator

The results of the mediation analysis reported in Table 3 confirm our assumption that information content is a channel through which internal governance improves liquidity. In Table 3 we use two model specifications; in the first model, we include only our variable of interest, internal governance, and other control variables. In the second model, we include both the mediator (information content) and internal governance. Based on the mediating analysis we expect a reduction in the size of the coefficient of our variable of interest, internal governance, and the mediator variable (information content) should be significant. The results show that when we include the information content in the model, the internal governance coefficients are reduced significantly for all 3 measures of liquidity and the information content variable is significant. To assess the significance of the reduction or mediation effect, we use Sobel's (1982) test. The results of the mediation analysis confirm our prediction that information content mediates the relationship between internal governance and stock market liquidity. The t-statistic from Sobel's test is statistically significant, suggesting that information content indeed serves as a channel through which internal governance affects stock market liquidity.

Insert Table 3

4.4. CEO Horizon, age, and Internal Governance

Acharya, Myers and Rajan (2011) argue that internal governance works effectively *only* when there is a real divergence between CEO's and the managers' appropriation horizon. To investigate the effectiveness of internal governance when a firm's CEO is close to retirement, we construct two subsamples: firms with CEOs above the mean CEO age (55 years old) and firms

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will be significant. The statistical test of mediation is given in the formula $t = {(\tau - \tau^*)} \Big/ {\sqrt{(\sigma_\tau^2 - \sigma_{\tau^*}^2 - 2\sigma_{\tau\tau^*}}}$, where σ_τ^2 is the variance of τ, $\sigma_{\tau^*}^2$ is the variance τ^* and $\sigma_{\tau\tau^*}$ is the covariance between τ and τ^*.

with CEOs with less than the mean age of 55 years old.[24] Table 4 Panel A presents summary statistics of CEOs' and subordinate managers' ages for the two subsamples. CEO above 55 years old subsample's statistics show that the mean CEO age is 59.90 years, while his subordinates have a mean age of 51.71 years. The mean relative age difference is 7.91 years which indicates that there is a significant difference in horizon between CEOs and their respective subordinates. On the contrary we find that the subsample with CEO less than 55 years old has no significant difference between CEO's age and subordinates managers' age.

Our regression results are reported in Table 4 Panel B. The regression results show that the coefficients on internal governance, as measured by age difference, are statistically significant for firms with CEOs close to retirement. However, the coefficients of internal governance measure in our second subsample, firms with CEOs below 55 years old, are statistically insignificant. All other corporate governance measures are still significant and have the consistent signs in both subsamples. These results support our Hypothesis 2 that the internal governance mechanism is effective in improving liquidity only when the remaining horizon of a CEO is short and subordinate managers have a longer horizon relative to the CEO.

Insert Table 4

Although the theoretical concept of internal governance in Acharya, Myers and Rajan (2011) is linked to the difference in horizon between the CEO and his immediate subordinates, the age level of the CEO, in addition to the age difference might also matter. For example, without controlling for CEO age we might equate an age difference between 95 and 91 where both CEO and his subordinates exceed the average retirement age with an age difference of 55 and 51 that shows a potential longer horizon for subordinates. In Table 5, we control for the CEO

[24] We test our hypothesis using mean CEO age of 55 years old (55 also happens to be the median) as a cutoff,

age to address this concern. The regression results confirm that internal governance, measured by age difference between the CEO and subordinate managers, increases stock market liquidity even after controlling for CEO age.

Insert Table 5

4.5. Firm-specific Knowledge and Internal Governance

Another key component for the effectiveness of internal governance is the level of firm-specific learning or effort the manager needs prior to becoming a CEO (Acharya, Myers, and Rajan, 2011). To test this argument, we adopt the excess value of human capital industrial rank proposed is Pantzalis and Park (2009) to differentiate between industries that require intensive human capital and those that are less dependent on human capital. We construct two subsamples of 24 industries each. The industry classification is based on Fama and French (1997) 48-industry listing. Table 6 Panel A reports the summary statistics for both subsamples. For the top 24 human capital value industries, we find that the average of CEO's age, subordinate managers' age, and relative age difference are 54.36 years, 50.56 years, and 3.67 years, respectively. Interestingly, we find a similar pattern for bottom 24 human capital value industries with average of CEO's age of 55.15 years, subordinate managers' age of 50.92 years, and relative age difference of 3.84 years.

Table 6, Panel B reports the results from regression analysis for the two subsamples of firms divided based on the value of human capital. We find a positive and statistically significant relation between liquidity and relative age difference only for firms in industries that have higher excess value of human capital. These results suggest that internal governance depends not only on the difference in horizons between the CEO and his subordinates, but also on the *nature of the job* and the relative importance of both the CEO and the subordinate managers within the firm.

In human capital intensive firms, subordinate managers have greater independence from their CEO, due to their knowledge and the higher cost of replacing them. Hence, internal governance is only effective for firms that require greater firm specific skills. Although the summary statistics show comparable statistics in terms of our internal governance measure, the regression analyses show that the nature of the job plays a vital role on the effectiveness of internal governance and internal governance is only effective in industries that require industry specific skills. These results support hypothesis 3 and the argument of Acharya, Myers and Rajan that internal governance is only effective in improving liquidity for firms that are dependent more on human capital for daily operations.

Insert Table 6

4.6. Subordinate Managers' Experience and Internal Governance

Prior accomplishments and experiences of subordinate managers enable them to effectively monitor CEO's myopic behavior. To examine the relationship between subordinate managers' performance and their ability to govern from within, we need to empirically measure subordinate managers' experience level. Antia, Pantzalis, and Park (2010) devise a measure of expected CEO decision horizon based on a combination of CEO tenure and age relative to the industry. We use their measure as a proxy for the relative experience of subordinate managers' in each firm to their peers in the same industry. As subordinate managers' knowledge and experience increase at firm and industry levels, their ability of effective internal monitoring also increases. The measure of subordinate managers' experience is defined as follows

$$Subordinate\ Managers'\ Experience = \left[Tenure_{i,t} - Tenure_{ind,t}\right] + \left[Age_{i,t} - Age_{ind,t}\right] \quad (7)$$

where $Tenure_{i,t}$, is the number of years that a subordinate manager has been with firm *i* and $Age_{i,t}$ is the age of the subordinate manager who works for firm *i* in year *t*. $Tenure_{ind,t}$ is the industry median of tenure and $Age_{ind,t}$ is the industry median of subordinate managers' ages. Industry classification is based on the Fama-French 48 industry classification. Given that the above measure is an industry-adjusted measure, it can take either positive or negative values. The comparison with other subordinate managers is conducted on two dimensions: the length of current tenure and age. The tenure reflects the firm-specific experience and knowledge, while the age reflects the accumulated experience beyond the executive position and outside the firm.

A positive value indicates that the subordinate manager's expected experience is more than the industry median either because the subordinate manager is older than the median age of other subordinate managers in the same industry and/or has been in his/her current position as long as or more than the industry median. Similarly, a negative value indicates that the expected experience is less than the industry median because the subordinate manager is younger and/or has been in his/her position for a shorter period of time than the median competitor firm's subordinate managers.

Table 7, Panel A reports that experienced subordinate managers have a mean age of 51.7 years, compared to 49.9 years for inexperienced managers. The relative age difference of experienced managers is 2.55 years compared to 4.19 years for inexperienced subordinate managers' sub-sample. Although experienced managers have shorter horizon than inexperienced managers, we expect to find that managers with more cumulative experience are capable of exerting better governance. Table 7, Panel B reports the regression results. We find that firms with more experienced subordinate managers are capable of implementing more effective internal monitoring than those who have subordinate managers with less experience than their industry peers. The coefficients of internal governance measure are statistically significant only for the experienced managers' subsample. These results support Hypothesis 4. Despite the fact that the summary statistics show that inexperienced subordinate managers have higher relative age difference, the regression results show that internal governance is only effective in improving liquidity for firms with experienced subordinate managers. These results highlight the importance of the accumulated experience for effective internal monitoring.

Insert Table 7

4.7. Controlling for the internal governance differences across industries

The nature of the industry maps the demographic attributes of its top management teams. Mature industries might have older top management team compared to industries in their infancy, such as internet and software industries. Emerging industries also might have CEOs with comparable ages as their subordinates. To account for variations in the top management team across different industries, we devise industry adjusted age difference measure of internal governance as described in equation 2.

Table 8 reports the results of regression analysis to rule out the possibility that our results are driven by industry variations. We find that the coefficients of industry adjusted internal governance measures are significantly negative for Gibbs estimate and percentage spread and significantly positive for turnover. These results are in line with the conjuncture that better internal governance leads to a higher stock market liquidity and are consistent with the results summarized in Table 2.

Insert Table 8

4.8. Positive vs. Negative Industry adjusted age difference

To further understand the relationship between the internal governance and stock market liquidity, we investigate if internal governance improves liquidity by dividing our sample into two subsamples with firms with positive and negative industry adjusted age differences. A positive (negative) value indicates that the age difference between the CEO and his subordinates for a given firm is greater than the industry mean age difference. Table 9, Panel A reports the descriptive statistics for both the subsamples. For positive industry adjusted age difference subsample, we find that the average CEO's age is 58.01 years and the average of subordinate managers' age is 49.23 years. The average of age difference between the CEO and his subordinates is almost 9 years, while the industry adjusted age difference is 5.25 years. On the other hand, the negative industry adjusted age difference subsample has mean CEO age of 51.38 years and the mean subordinate managers' age of 52.07 years. The average age difference between the CEO and his subordinates is -1.35 years, while the industry adjusted age difference is -5.11 years.

Table 9, Panel B summarizes the regression results for positive and negative industry adjusted age difference subsamples. We find that only the firms with a positive age difference relative to operating industry have significant coefficients of internal governance as measured by the industry adjusted age difference. The coefficient of industry adjusted internal governance measure for turnover is positive and statistically significant. Also, the coefficients of both percentage spread and Gibbs estimate are negative and statistically significant. These results are consistent with our earlier findings that firms with higher internal governance are more liquid. However, for the firms with negative industry adjusted age difference, the level of internal governance does not significantly influence the liquidity.

Insert Table 9

5. Alternative internal governance measures

Given the difficulty of measuring internal governance motivated by the difference in horizons, age difference between the CEO and his subordinates may not fully capture this multi-dimensional concept. To alleviate this concern, we develop alternative measures that capture the essence of internal governance from different perspectives.

5.1. Stock Options Exercise

The executives' compensation can also be related to the executives' horizon in the firm and their expectation about the future performance (Carpenter and Remmers, 2001, Aboody, Hughes, Liu and Su, 2008, and Brook, Chance and Cline, 2012). In particular, previous literature links executives' stock options exercise to operating performance, executives' horizon, and possession of private information. Brook, Chance, and Cline (2012) show that executives tend to exercise more stock options around retirement or resignation from the firm.

With unexercised exercisable options an executive has the choice to reduce their equity holdings (Goplan et al., 2014) if they have a shorter perceived horizon within the firm. In this case the executive might undertake myopic actions that boost short-term performance and leave the firm before the long-term cost of their decisions is realized. On the other hand, if an executive decides to accumulate their equity holdings over time, it might reflect a long term horizon of the executive. To proxy an executive's horizon in a firm, we calculate the ratio of top executive's exercisable unexercised stock options value to the executive's total pay. If an executive has a short horizon within the firm, he would probably exercise a larger proportion of exercisable options, if not all of them, than if he expects a strong long term performance. As the ratio of exercisable unexercised options of subordinate managers to CEO increases, the subordinates' horizon is more closely tied to the long term performance of the firm.

$$Executive's\ stock\ options_{\ i,t} = \frac{Exerciable\ unexercised\ option\ value_{i,t}}{TOTAL\ PAY_{i,t}} \tag{8}$$

$$Options\ ratio_{\ i,t} = \frac{Average\ subordinates'percentage\ of\ options_{\ i,t}}{Percentage\ of\ CEO's\ option_{\ i,t}} \tag{9}$$

where, $Exercisable\ unexercised\ stock\ option\ value_{i,t}$ is the value of executive's unexercised exercisable stock options, $TOTAL\ PAY_{i,t}$ is the executive's total pay, $Average\ subordinates' percentage\ of stock\ options_{\ i,t}$ is the mean of unexercised but exercisable options percentage of non-CEO executives, and $Percentage\ of\ CEO's\ option_{\ i,t}$ is the ratio of unexercised exercisable options to the CEO's total pay.

Table 10 shows the regression results for options ratio. The results suggest that as ratio of subordinate managers' unexercised exercisable options to the CEO increases, the liquidity of the firm also increases. In other words, as the horizon of subordinate managers increases, the liquidity of the firm tends to increase. The coefficients of options ratio are statistically significant for all the liquidity measures. The results from this alternative proxy of relative horizon confirm hypothesis 1 and are in line with the results reported in Table 2.

Insert Table 10

5.2. The CEO Pay Slice

According to Acharya, Myers and Rajan (2011), internal governance is effective only if both the CEO and the subordinate managers contribute to the firm. If CEO dominates power, or contribution to the firm, internal governance will not be effective as the subordinate managers' won't be able to influence the CEO's decisions. Based on this argument and following Bebchuk, Cremers, and Peyer (2011), we use CEO pay slice as an alternative measure of internal governance.

Bebchuk, Cremers and Peyer (2011) examine the impact of CEO dominance and centrality on the firm performance. They use the fraction of CEO's aggregate compensation to top executives' aggregate compensation as a proxy of CEO dominance and find that CEO dominance is positively associated with higher agency problem and negatively associated with the firm value. The fraction of CEO compensation to other executives' pay reflects the relative importance of the CEO as well as the extent to which the CEO is able to extract rents. We use the CEO pay slice to the total pay of top 5 executive managers (the CEO and top 4 subordinate managers), as an inverse measure of internal governance and the relative subordinates' power:

$$CEO\ Pay\ Slice_{i,t} = \frac{CEO\ PAY_{i,t}}{TOP\ 5\ TOTAL\ PAY_{i,t}} \tag{10}$$

where $CEO\ PAY_{i,t}$ is the CEO's total compensation, including salary, bonus, other annual pay, the total value of restricted stock granted that year, the Black and Scholes value of stock options granted that year, long- term incentive payouts, and all other total compensation (as reported in ExecuComp item TDC1). $TOP\ 5\ TOTAL\ PAY_{i,t}$ is the total compensation of the top 5 executives including the CEO. Higher CEO power indicates a lower or no influence of subordinate managers on the CEO and hence, weaker internal governance.

Using the CEO pay slice, we re-evaluate the impact of internal governance on liquidity. Table 11 summarizes the results from this additional analysis. We find that CEO Pay Slice is negatively related to liquidity, suggesting that the firms with higher subordinate power are more liquid than other firms. The results are consistent with our Hypothesis 1, higher internal governance leads to higher liquidity, and serve as a robustness check for our main findings in Table 2 using age difference as a proxy of internal governance. The results also support Bebchuk, Cremers, and Peyer (2011) findings that CEO dominance increases the agency problem.

Insert Table 11

6. Additional Analyses

6.1. Concentrated ownership (family firms)

Acharya, Myers, and Rajan's (2011) model assumes that the CEO and subordinate managers have divergent horizons in the firm. However, the difference in horizons and inefficient use of resources may not be relevant to firms that are owned and managed by family members. In family controlled firms, the CEO and his subordinates, presumably, share the same objectives and do not have the divergent misappropriation horizon problem. As we have no access to data on family ownership, we use the ownership concentration of the top executive in the firm as a proxy for family controlled firms. Untabulated results indicate that internal governance is not effective in enhancing stock market liquidity for firms with executive ownership greater than 10 percent. However, the relationship between stock market liquidity and internal governance is statistically significant for the rest of the sample.[25] This lends support to the assertion that internal governance, as a monitoring mechanism, is less critical to mitigate agency problems when CEO and firm's interests are aligned, which might be the case with firms with concentrated ownership.

6.2. Endogeneity and causality

Although we find that the internal governance, regardless of how it is measured, is positively related to stock market liquidity, the OLS regressions may not fully account for the potential endogeneity and reversed causality in the sample. Modeling the relationship between governance and stock market liquidity may be problematic if there is an endogenous feedback from stock market liquidity to internal governance because liquidity and governance are jointly determined. Though not specifically related to our internal governance measure, many voice or exit theories model the dependence of governance on liquidity. Specifically a strand of literature on blockholders suggests that liquidity strengthens governance as it facilitates formation of a

[25] Our results are robust to alternative cutoffs for executive ownership such as 15% and 30%.

block (Edmans, 2009) and the blockholder to gather more information and trade more aggressively once a block is formed (Admati and Pfleiderer, 2009; Edmans, 2009; Edmans and Manso, 2010). Different from theories that suggest liquidity has a causal effect on governance provided by blockholders, it is difficult to reason that liquidity such as spread and turnover can directly affect the age difference between the CEO and his subordinates. Thus, the main results of this paper may not be subject to serious reverse causality issues.

Nevertheless, to address this issue, we analyze the relation between changes in liquidity and changes in internal governance (our main measure is age difference). If changes in internal governance have a significant influence on changes in liquidity, then as internal governance strengthens over time, we would expect to see corresponding increases in liquidity. Regressions with changes in variables used in Aggarwal, Erel, Ferreira and Matos (2011) and Chung, Elder and Kim (2010) are considered advantageous as they are generally less likely to show spurious relations between the variables than regressions using the level variables. Year-to-year changes in variables provide a stronger test of causal relations than do levels of these variables because the levels of many variables can be cross-sectionally correlated without any direct causal link. While correlations in changes do not imply causality either, a failure to find correlation in changes is likely to indicate no causal relation. The results (Table 12) show that the coefficient on the changes in internal governance is significantly positively related to changes in liquidity indicating that an increase in liquidity tends to be associated with an increase in the firm's internal governance measure.

Insert Table 12

In addition, following Aggarwal et al (2011), we also run the changes regression analysis in the reverse direction, using the change in liquidity as the explanatory variable and the change in internal governance as the dependent to determine whether higher liquidity drives improvements in internal governance. We estimate three different models using changes in turnover, spread, and Gibbs estimate respectively. Results for the reverse changes regressions (not reported for brevity), reports a statistically insignificant coefficient for the change in liquidity. Overall, our results are not consistent with alternative stories which emphasize the reverse causality from liquidity to governance.

To further support our inference about the relationship between internal governance and stock market liquidity, we present a "quasi" experiment to confirm our results. We created a dummy variable to detect changes in top management team that increases the age difference between CEO and his subordinate managers. Positive changes in the age difference can be triggered either due to the replacement of existing subordinate manager(s) with younger manager(s) or due to replacement of the existing CEO with an older CEO or both. In such events the internal governance, measured by age difference between the CEO and his subordinates, should be stronger resulting in a higher stock market liquidity. Results presented in Table 13 report a statistically significant and negative coefficient for *age difference increase* dummy for spread and Gibbs estimate. These results confirm the central theme of the paper and establish a credible causal inference between internal governance and stock market liquidity.

Insert Table 13

6.3. Fixed Effect Estimation

In this section, we check for the robustness of our results with respect to different estimation methods. In order to improve estimation efficiency, we analyze the relationship between stock market liquidity and the different levels of monitoring using industry fixed effect. The fixed effect controls for the industry's specific characteristics that are not captured by other control variables. Results reported in Table 14 indicate that the positive relationship between internal governance and liquidity is robust to the alternate model specifications.

Insert Table 14

7. Conclusion

We examine the impact of internal governance on the stock market liquidity for a sample of S&P 1500 constituent firms. Acharya, Myers, and Rajan (2011) introduce a theoretical model of internal governance based on partnership between the current CEO and his subordinates who have different horizons. In order to protect their future in the firm, subordinate managers need to effectively exert internal governance to assure that the CEO makes firm value maximizing decisions. Their efforts are expected to improve operating and financial transparency of the firm and result in a better information environment and higher liquidity.

Following Acharya, Myers, and Rajan (2011) we use a measure of internal corporate governance that captures the divergence of appropriation horizons between the current CEO and his subordinate managers to empirically test the relationship between internal governance and liquidity. We proxy this divergence of horizons by the mean relative age difference between the CEO's age and the top four subordinate managers' ages. As there is an inherent limitation with using the relative age difference as a proxy for internal governance, we develop an adjusted age difference measure to control for other factors such as industry differences that are not associated with the appropriation horizon. In addition, we use several measures of horizons from the executive compensation literature, such as the unexercised exercisable options and CEO pay slice, as alternative measures of internal governance.

Our results document that firms with a larger divergence of appropriation horizons are more liquid. We further show that internal monitoring is more effective in improving liquidity for firms with CEOs with relatively shorter horizon, with more experienced subordinate managers, and that require a higher degree of firm specific knowledge and skills. To further strengthen our analysis, we test whether information content is the channel through which internal governance affects liquidity. Using Amihud price impact as a proxy of information content (Brennan, Huh and Subrahmanyam, 2013; Amihud, 2002) and Sobel's (1982) test of mediation, we show that internal governance significantly increase the informativeness of stock prices and information content indeed serves as a channel through which internal governance can affect stock market liquidity.

Our results are consistent with the argument that internal managers are the most important source of firm specific information within the firm (Fama and Jensen, 1983; Core, Holthausen and Larcker, 1999; Masulis and Mobbs, 2011). Powerful subordinates have the ability to elicit better decision-making of CEOs and help the CEO and the firm to disclose more reliable information to the market. Firms with such internal governance can effectively reduce information asymmetry and hence improve its liquidity.

We document that the positive relationship between internal governance and liquidity is robust to inclusion of other governance mechanisms. We include three widely accepted corporate governance measures (governance index, institutional ownership, and number of analysts following a firm) as control variables. Consistent with Chung, Elder, and Kim (2010) and Chung and Zhang (2011), we find that governance index is positively related to liquidity. We also document that institutional investors prefer stocks of better-governed firms for liquidity reasons and analysts follow stocks with lower liquidity (Chung, Elder, and Kim, 2010; Van Ness, Van Ness, and Warr, 2001; Chung, McInish, Wood, and Wyhowski, 1995). Most importantly, our internal governance measure remains a significant factor in determining liquidity of a firm even after controlling for other forms of corporate governance.

In addition, we find that changes in our internal governance measures are significantly related to changes in internal governance measures over time, suggesting that firms may improve stock market liquidity by focusing not only on the role of CEO, but also on optimally configuring the executive suite to include subordinates with longer horizons and have the ability and incentives to exert monitoring from bottom-up. In sum, internal governance is positively related to liquidity and our results are robust to inclusion of conventional governance measures, alternative model specifications, and different measures of internal monitoring and liquidity.

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Table 1
Descriptive Statistics

Table 1 presents descriptive statistics for the sample firms for the period from 1 January 2001 to 31 December 2007. *CEO Age* is the CEO's age measured in years, *Subordinates Managers' Age* is the mean age of the top 4 subordinate managers (non-CEO executives) in years, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinates managers' Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index ranges from 0 to 24, *Institutional Ownership* is the proportion of outstanding stocks held by Institutional Investors, *Analyst Coverage* is the mean number of analysts following a firm, *Turnover* is average daily volume divided by total shares outstanding, *Percentage Spread* is the dollar spread divided by the midpoint of the bid and ask prices, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Closing Price* is the mean daily stock price at the market close, *Volatility* is the annualized standard deviation of daily returns, *Trading Volume* is the mean daily dollar trading volume, *Total Assets* is the book value of a firm's total assets, *Intangible Assets* is the book value of total intangible assets, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Dividend* is the dividend per year, and *Market Return* is the annualized daily return on S&P 500 index.

Variable	N	Mean	Standard Deviation	Minimum	Maximum
			Internal Monitoring		
CEO Age	6,539	54.68	6.25	34.00	91.00
Subordinates managers' age	6,539	50.69	4.53	33.00	81.00
Age difference	6,539	3.75	6.96	-25.00	36.33
Option's ratio	6,539	1.74	12.53	0	20.48
CEO Pay Slice	6,539	0.40	0.12	0.01	1.00
			Internal/ External Monitoring		
Gov-Index	4,579	12.49	2.86	3.00	21.00
External Monitoring					
Institutional Ownership	5,547	0.71	0.17	0.00	1.00
Analyst Coverage	6,494	10.87	6.55	1.00	31.92
			Liquidity and Information Content Measures		
Turnover (%)	5,688	0.26	0.16	0.04	1.70
Percentage Spread (%)	6,539	0.25	0.36	0.02	3.31
Gibbs estimate (%)	6,539	0.66	0.46	0.02	4.74
Amihud's information content	5254	1.19	0.57	0.29	6.16
			Control Variables		
Closing Price ($)	6,539	38.32	22.61	5.04	463.95
Volatility % (Annual)	6,538	30.28	12.56	8.91	155.24
Trading Volume (Millions)	6,539	58.09	118.07	0.04	2,036.33
Total Assets (Millions)	6,324	16,665	83,814	0.01	2,187,631
Intangible Assets (Millions)	6,021	2,086	7,525	0.00	129,115
R & D Expenditure (Millions)	5,456	7.28	121.10	0.00	5,052.00
Dividend ($)	6,175	0.53	0.83	0.00	17.27
Market Return (%) (Annual)	6,539	2.74	20.66	-114.07	75.07

Table 2
Internal Governance and stock market liquidity

This table reports the regression analysis of Stock market liquidity on internal governance using the following regression model:

Turnover$_{i,t}$ *or Percentage Spread*$_{i,t}$ *or Gibbs estimate*$_{i,t}$ = β_0 + β_1 *Internal Governance* (*Age Difference)*$_{i,t}$ + β_2 *Gov-Index*$_{i,t}$ + β_3 *Institutional Ownership*$_{i,t}$ + β_4 *Analyst Coverage*$_{i,t}$ +β_5 *Log(Closing Price*$_{i,t}$*)* + β_6 *R&D Expenditure*$_{i,t}$ + β_7 *Total Assets*$_{i,t}$ + β_8 *Log(Trading Volume*$_{i,t}$*)* + β_9 *Volatility*$_{i,t}$+ β_{10} *Intangible Asset*$_{i,t}$+ β_{11} *Dividend*$_{i,t}$ + β_{12} *Market Return*$_{i,t}$ + $\varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index , *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Turnover			Percentage Spread			Gibbs estimate		
	Model l	Model2	Model3	Model1	Model2	Model3	Model1	Model2	Model3
Internal Governance	0.030**	0.034***	0.036***	-0.027**	-0.030**	-0.037***	-0.029**	-0.028**	-0.029**
(Age difference)	(2.17)	(2.65)	(2.85)	(-2.14)	(-2.46)	(-3.07)	(-2.13)	(-2.07)	(-2.15)
Gov-Index			0.067***			-0.176***			-0.053***
			(4.83)			(-13.46)			(-3.62)
Institutional		0.275***	0.265***		-0.159***	-0.134***		-0.016	-0.009
Ownership		(20.67)	(19.80)		(-12.42)	(-10.61)		(-1.13)	(-0.62)
Analyst Coverage		-0.139***	-0.126***		0.220***	0.188***		0.089***	0.080***
		(-7.13)	(-6.47)		(11.76)	(10.21)		(4.39)	(3.90)
Closing Price	0.137***	0.079***	0.090***	-0.117***	-0.057***	-0.087***	-0.077***	-0.059***	-0.069***
	(7.62)	(4.55)	(5.18)	(-6.88)	(-3.40)	(-5.29)	(-4.31)	(-3.26)	(-3.74)
R& D Expenditure	0.025*	0.014	0.013	-0.013	-0.003	0.001	0.008	0.011	0.012
	(1.77)	(1.10)	(0.97)	(-0.97)	(-0.25)	(0.11)	(0.55)	(0.78)	(0.88)
Total Assets	-0.066***	-0.046***	-0.046***	0.020	0.009	0.011	0.032**	0.031**	0.031**
	(-4.31)	(-3.19)	(-3.25)	(1.39)	(0.65)	(0.81)	(2.05)	(1.96)	(2.02)
Trading Volume	0.418***	0.485***	0.453***	-0.210***	-0.367***	-0.285***	-0.177***	-0.248***	-0.223***
	(23.27)	(20.59)	(18.63)	(-12.45)	(-16.19)	(-12.42)	(-9.96)	(-10.07)	(-8.71)
Volatility	0.448***	0.424***	0.435***	0.375***	0.405***	0.376***	0.517***	0.528***	0.520***
	(28.23)	(28.21)	(28.71)	(25.09)	(28.01)	(26.35)	(32.90)	(33.49)	(32.59)
Intangible assets	-0.154***	-0.123***	-0.123***	0.058***	0.043***	0.044***	0.004	0.005	0.005
	(-9.44)	(-7.99)	(-8.05)	(3.77)	(2.91)	(3.06)	(0.22)	(0.32)	(0.28)
Dividend	-0.028**	-0.006	-0.011	0.019	0.010	0.023*	-0.058***	-0.058***	-0.054***
	(-2.00)	(-0.43)	(-0.80)	(1.42)	(0.74)	(1.81)	(-4.07)	(-4.09)	- (-3.78)
Market Return	0.107***	0.084***	0.082***	-0.262***	-0.242***	-0.237***	-0.046***	-0.038***	-0.037***
	(7.55)	(6.29)	(6.15)	(-19.70)	(-18.90)	(-18.89)	(-3.30)	(-2.75)	(-2.62)
Adjusted R^2	0.328	0.409	0.413	0.404	0.453	0.479	0.422	0.428	0.43

Table 3
Mediation Analysis: Internal Governance, Information Asymmetry, and Stock market liquidity

This table reports the mediating regression analysis to assess whether the information asymmetry serves as a channel through which internal governance affects liquidity:

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (Age\ Difference)_{i,t} + \beta_2\ Information\ Asymmetry_{i,t} + \beta_3\ Gov\text{-}Index_{i,t} + \beta_4\ Institutional\ Ownership_{i,t} + \beta_5\ Analyst\ Coverage_{i,t} + \beta_6\ Log(Closing\ Price_{i,t}) + \beta_7\ R\&D\ Expenditure_{i,t} + \beta_8\ Total\ Assets_{i,t} + \beta_9\ Log(Trading\ Volume_{i,t}) + \beta_{10}\ Volatility_{i,t} + \beta_{11} Intangible\ Asset_{i,t} + \beta_{12}\ Dividend_{i,t} + \beta_{13}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Information Asymmetry* $_{i,t}$ is the turnover version of Amihud illiquidity and is calculated as explained in equation 9, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. To assess the significance of the reduction mediation effect, we use Sobel test following Sobel (1982). Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

	Turnover		Percentage Spread		Gibbs estimate	
Internal Governance	0.037***	0.026**	-0.036***	-0.027**	-0.031**	-0.029**
(Age difference)	(2.90)	(2.25)	(-3.00)	(-2.39)	(-2.28)	(-2.20)
Information Asymmetry		-0.366***		0.327***		0.141***
		(-25.53)		(23.66)		(8.28)
Gov-Index	0.065***	0.044***	-0.168***	-0.150***	-0.051***	-0.040***
	(4.71)	(3.47)	(-12.81)	(-12.19)	(-3.47)	(-2.79)
Institutional Ownership	0.262***	0.178***	-0.127***	-0.051***	-0.007	0.031**
	(19.78)	(14.05)	(-10.00)	(-4.19)	(-0.52)	(2.08)
Analyst Coverage	-0.130***	-0.069***	0.197***	0.142***	0.088***	0.064***
	(-6.72)	(-3.85)	(10.65)	(8.20)	(4.30)	(3.16)
Closing Price	0.084***	0.094***	-0.069***	-0.078***	-0.067***	-0.071***
	(4.90)	(5.93)	(-4.20)	(-5.07)	(-3.69)	(-3.91)
R&D Expenditure	0.013	0.008	0.001	0.005	0.013	0.014
	(1.01)	(0.70)	(0.08)	(0.45)	(0.97)	(1.05)
Total Assets	-0.047***	-0.034***	0.011	0.000	0.029*	0.026*
	(-3.32)	(-2.62)	(0.85)	(0.00)	(1.90)	(1.70)
Trading Volume	0.469***	0.256***	-0.318***	-0.127***	-0.229***	-0.144***
	(19.50)	(10.79)	(-13.86)	(-5.57)	(-8.98)	(-5.28)
Volatility	0.432***	0.472***	0.050***	0.011	0.008	-0.012
	(28.80)	(33.89)	(3.42)	(0.82)	(0.45)	(-0.71)
Intangible assets	-0.125***	-0.082***	0.374***	0.338***	0.519***	0.502***
	(-8.25)	(-5.84)	(26.16)	(25.21)	(32.64)	(31.59)
Dividend	-0.010	-0.023*	0.018	0.030**	-0.053***	-0.048***
	(-0.75)	(-1.91)	(1.42)	(2.54)	(-3.74)	(-3.44)
Market Return	0.079***	0.057***	-0.226***	-0.206***	-0.032**	-0.024*
	(5.98)	(4.67)	(-17.96)	(-17.54)	(-2.31)	(-1.70)
	Sobel test	**2.126****	***Sobel test***	**2.124****	***Sobel test***	**0.691**
Adjusted R^2	0.413	0.500	0.465	0.535	0.422	0.434

Table 4
Internal governance, CEO Horizon and Stock market liquidity

This table reports the descriptive statistics and regression analysis for the impact of CEO horizon on the effectiveness of Internal Governance. To investigate this impact, we divide the sample firms into two subsamples; firms with CEOs above the sample mean CEO age (55 years old) and firms with CEOs below the sample mean CEO age and estimate the following regression model:

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (age\ difference)_{i,t} + \beta_2\ Gov\text{-}Index_{i,t} + \beta_3\ Institutional\ Ownership_{i,t} + \beta_4\ Analyst\ Coverage_{i,t} + \beta_5\ Log(Closing\ Price_{i,t}) + \beta_6\ R\&D\ Expenditure_{i,t} + \beta_7\ Total\ Assets_{i,t} + \beta_8\ Log(Trading\ Volume_{i,t}) + \beta_9\ Volatility_{i,t} + \beta_{10}\ Intangible\ Asset_{i,t} + \beta_{11}\ Dividend_{i,t} + \beta_{12}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* i the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Panel A: Descriptive statistics: Internal Governance for the sub-samples of firms divided based on CEO horizon

Variable	N	Mean	Median	Minimum	Maximum	SD
CEO above the mean CEO's age						
CEO Age	2077	59.90	59.00	55.33	91.00	4.31
Subordinate Managers' Age	2077	51.71	51.67	36.00	81.00	4.47
Internal Governance (Age difference)	2050	7.91	7.33	-25.00	36.33	6.11
CEO under the mean CEO's age						
CEO Age	2261	49.85	51.00	34.00	54.80	3.70
Subordinate Managers' Age	2261	49.75	49.67	33.00	76.00	4.39
Internal Governance (Age difference)	2228	-0.08	0.25	-22.00	16.00	5.52

Panel B. Summary of regression results						
Variable	CEO above the mean CEO's age			CEO under the mean CEO's age		
	Turnover	Percentage Spread	Gibbs estimate	Turnover	Percentage Spread	Gibbs estimate
Internal Governance	0.065***	-0.061***	-0.065***	-0.001	0.018	-0.019
(Age difference)	(3.62)	(-3.46)	(-3.31)	(-0.05)	(1.08)	(-1.02)
Gov-Index	0.045**	-0.182***	-0.091***	0.087***	-0.167***	-0.011
	(2.35)	(-9.55)	(-4.29)	(4.42)	(-9.34)	(-0.55)
Institutional Ownership	0.285***	-0.145***	0.007	0.236***	-0.123***	-0.025
	(15.51)	(-7.93)	(0.35)	(12.18)	(-6.98)	(-1.24)
Analyst Coverage	-0.114***	0.144***	0.050	-0.133***	0.214***	0.117***
	(-4.06)	(5.17)	(1.64)	(-4.95)	(8.75)	(4.25)
Closing Price	0.102***	-0.089***	-0.085***	0.075***	-0.085***	-0.051**
	(4.15)	(-3.68)	(-3.17)	(3.06)	(-3.78)	(-2.03)
R&D Expenditure	0.001	0.006	0.012	0.016	-0.001	0.017
	(0.05)	(0.32)	(0.59)	(0.85)	(-0.08)	(0.87)
Total Assets	-0.045**	0.008	0.050**	-0.123***	0.032	-0.017
	(-2.27)	(0.42)	(2.16)	(-5.64)	(1.62)	(-0.76)
Trading Volume	0.484***	-0.231***	-0.156***	0.457***	-0.335***	-0.297***
	(13.82)	(-6.65)	(-4.14)	(13.56)	(-10.92)	(-8.52)
Volatility	0.484***	0.389***	0.510***	0.385***	0.372***	0.536***
	(23.19)	(18.78)	(22.11)	(17.74)	(18.83)	(24.18)
Intangible assets	-0.120***	0.036*	-0.024	-0.100***	0.047**	0.046*
	(-5.63)	(1.72)	(-0.96)	(-4.48)	(2.30)	(1.94)
Dividend	-0.017	0.013	-0.044**	-0.002	0.028	-0.071***
	(-0.93)	(0.73)	(-2.17)	(-0.11)	(1.56)	(-3.48)
Market Return	0.083***	-0.216***	-0.019	0.078***	-0.251***	-0.055***
	(4.53)	(-11.94)	(-0.95)	(4.11)	(-14.49)	(-2.81)
Adjusted R^2	0.460	0.469	0.417	0.389	0.495	0.456

Table 5

Internal governance, CEO age and stock market liquidity:

To control the level of CEO age, we analyze the following regression model:

Turnover$_{i,t}$ *or Percentage Spread*$_{i,t}$ *or Gibbs estimate*$_{i,t}$ = β_0 + β_1 *Internal Governance (age difference)*$_{i,t}$ + β_2 *CEO Age*$_{i,t}$ + β_3 *Gov-Index*$_{i,t}$ + β_4 *Institutional Ownership*$_{i,t}$ + β_5 *Analyst Coverage*$_{i,t}$ +β_6 *Log(Closing Price*$_{i,t}$*)* + β_7 *R&D Expenditure*$_{i,t}$ + β_8 *Total Assets*$_{i,t}$ + β_9 *Log(Trading Volume*$_{i,t}$*)* + β_{10} *Volatility*$_{i,t}$+ β_{11} *Intangible Asset*$_{i,t}$ + β_{12} *Dividend*$_{i,t}$ + β_{13} *Market Return*$_{i,t}$ + $\varepsilon_{i,t}$

Turnover is average daily volume as a proportion of shares outstanding, *Percentage Spread*, which is the percent quoted spread, *Gibbs estimate*, which is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *CEO Age* is the CEO age, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, *Industry* is a dummy variable to control for industry fixed effects, *Year* is a dummy variable to control for time fixed effects, and $\varepsilon_{i,t}$ is the error term. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Turnover	Percentage Spread	Gibbs estimate
Internal Governance (Age difference)	0.047**	-0.009	-0.052***
	(2.66)	(-0.43)	(-2.44)
CEO Age	-0.014	-0.038**	0.031
	(-0.76)	(-2.03)	(1.56)
Gov-Index	0.067***	-0.175***	-0.053***
	(4.72)	(-12.89)	(-3.57)
Institutional Ownership	0.265***	-0.135***	-0.008
	(19.77)	(-9.88)	(-0.59)
Analyst Coverage	-0.127***	0.185***	0.081***
	(-5.61)	(10.23)	(4.33)
Closing Price	0.091***	-0.086***	-0.069***
	(4.2)	(-4.88)	(-3.97)
R&D Expenditure	0.012	0.001	0.012
	(0.89)	(0.14)	(2.2)
Total Assets	-0.046***	0.012	0.031**
	(-2.45)	(1.84)	(0.69)
Trading Volume	0.454***	-0.283***	-0.224***
	(16.37)	(-10.91)	(-7.79)
Volatility	0.434***	0.375***	0.521***
	(20.2)	(18.28)	(22.57)
Intangible assets	-0.123***	0.045***	0.004
	(-6.06)	(4.00)	(0.15)
Dividend	-0.011	0.022*	-0.053***
	(-0.87)	(1.96)	(-4.59)
Market Return	0.082***	-0.237***	-0.036***
	(6.59)	(-17.00)	(-2.26)
Adjusted R^2	0.4108	0.4775	0.4282

Table 6
Internal governance, Human capital, and Stock market liquidity

This table reports the descriptive statistics and regression analysis of the impact of firm specific learning (human capital) on the effectiveness of Internal Governance. To explore this relation, we use Pantzalis and Park (2009) rank of the excess value of human capital for each industry to divide our sample firms into top 24 human capital industries and the bottom 24 human capital industries. We estimate the following regression model for each group:

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (age\ difference)_{i,t} + \beta_2\ Gov\text{-}Index_{i,t} + \beta_3\ Institutional\ Ownership_{i,t} + \beta_4\ Analyst\ Coverage_{i,t} + \beta_5\ Log(Closing\ Price_{i,t}) + \beta_6\ R\&D\ Expenditure_{i,t} + \beta_7\ Total\ Assets_{i,t} + \beta_8\ Log(Trading\ Volume_{i,t}) + \beta_9\ Volatility_{i,t} + \beta_{10}\ Intangible\ Asset_{i,t} + \beta_{11}\ Dividend_{i,t} + \beta_{12}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Panel A: Descriptive statistics: Internal Governance for the sub-samples of firms divided based on firm-specific knowledge

Variable	N	Mean	Median	Minimum	Maximum	SD
Top Human Capital value Industries						
CEO Age	2750	54.36	54.50	34.00	91.00	6.38
Subordinate Managers Age	2750	50.56	50.65	33.00	70.00	4.48
Internal Governance (Age difference)	2713	3.67	3.67	-18.50	29.33	6.80
Bottom Human Capital value Industries						
CEO Age	1816	55.15	55.00	34.00	87.00	5.98
Subordinate Managers Age	1816	50.92	50.85	36.00	81.00	4.59
Internal Governance (Age difference)	1787	3.84	3.50	-25.00	36.33	7.16

Panel B. Summary of regression results						
Variable	Top 24 Industries			Bottom 24 Industries		
	Turnover	Percentage Spread	Gibbs estimate	Turnover	Percentage Spread	Gibbs estimate
Internal Governance (Age difference)	0.046***	-0.048***	-0.032*	0.025	-0.031	-0.024
	(2.58)	(-2.98)	(-1.75)	(1.37)	(-1.75)	(-1.16)
Gov-Index	0.055***	-0.194***	-0.034*	0.085***	-0.158***	-0.073***
	(2.78)	(-10.77)	(-1.72)	(4.44)	(-8.26)	(-3.35)
Institutional Ownership	0.244***	-0.115***	-0.006	0.266***	-0.151***	-0.011
	(13.00)	(-6.74)	(-0.34)	(13.89)	(-7.90)	(-0.50)
Analyst Coverage	-0.133***	0.167***	0.095***	-0.131***	0.223***	0.067**
	(-4.91)	(6.78)	(3.50)	(-4.67)	(8.00)	(2.13)
Closing Price	0.070***	-0.063***	-0.045*	0.103***	-0.121***	-0.096***
	(2.90)	(-2.87)	(-1.86)	(4.09)	(-4.84)	(-3.37)
R&D Expenditure	0.018	-0.003	0.014	0.028	-0.015	0.011
	(0.99)	(-0.18)	(0.71)	(1.55)	(-0.84)	(0.54)
Total Assets	-0.055***	0.018	0.044**	-0.104***	-0.002	-0.005
	(-2.75)	(0.97)	(2.03)	(-4.55)	(-0.07)	(-0.21)
Trading Volume	0.448***	-0.272***	-0.238***	0.521***	-0.302***	-0.212***
	(13.25)	(-8.83)	(-6.94)	(14.52)	(-8.44)	(-5.28)
Volatility	0.391***	0.405***	0.558***	0.498***	0.348***	0.468***
	(18.32)	(20.81)	(25.80)	(23.21)	(16.30)	(19.28)
Intangible assets	-0.141***	0.048**	0.009	-0.068***	0.052**	0.006
	(-6.47)	(2.42)	(0.38)	(-3.06)	(2.34)	(0.25)
Dividend	-0.007	0.034**	-0.067***	-0.018	0.011	-0.031
	(-0.39)	(1.97)	(-3.45)	(-0.99)	(0.59)	(-1.48)
Market Return	0.049***	-0.235***	-0.048**	0.126***	-0.239***	-0.024
	(2.66)	(-13.90)	(-2.54)	(6.72)	(-12.81)	(-1.15)
Adjusted R^2	0.373	0.479	0.444	0.483	0.487	0.414

Table 7
Internal governance, Subordinate managers' experience and Stock market liquidity

This table reports the descriptive statistics and regression analysis of the impact of subordinate managers' experience, relative to their peers in the same industry, on liquidity. We divide our sample firms into two sub-samples based on Antia, Pantzalis and Park (2010) measure of expected decision horizon. We estimate the following regression model for each of the sub-sample:

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (age\ difference)_{i,t} + \beta_2\ Gov\text{-}Index_{i,t} + \beta_3\ Institutional\ Ownership_{i,t} + \beta_4\ Analyst\ Coverage_{i,t} + \beta_5\ Log(Closing\ Price_{i,t}) + \beta_6\ R\&D\ Expenditure_{i,t} + \beta_7\ Total\ Assets_{i,t} + \beta_8\ Log(Trading\ Volume_{i,t}) + \beta_9\ Volatility_{i,t} + \beta_{10}\ Intangible\ Asset_{i,t} + \beta_{11}\ Dividend_{i,t} + \beta_{12}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Panel A: Internal Governance for the sub-samples of firms divided based on Subordinate Managers level of Experience						
Variable	N	Mean	Median	Minimum	Maximum	SD
Experienced Subordinates Managers						
CEO Age	503	54.64	54.00	39.00	91.00	6.69
Subordinate Managers Age	503	51.71	51.50	39.00	79.00	5.40
Internal Governance (Age difference)	500	2.55	2.83	-25.00	23.67	7.70
Inexperienced Subordinates Managers						
CEO Age	532	54.28	54.50	38.00	78.00	6.38
Subordinate Managers Age	532	49.87	49.71	39.33	81.00	4.28
Internal Governance (Age difference)	526	4.19	4.21	-25.00	27.67	7.04

Panel B. Summary of regression results						
Variable	Experienced Subordinate Managers			Inexperienced Sub Managers		
	Turnover	Percentage Spread	Gibbs estimate	Turnover	Percentage Spread	Gibbs estimate
Internal Governance (Age difference)	0.134***	-0.162***	-0.048	0.012	-0.050	-0.038
	(3.87)	(-4.56)	(-1.21)	(0.36)	(-1.58)	(-0.96)
Gov-Index	0.038	-0.113***	0.000	0.000	-0.084**	-0.099**
	(0.99)	(-2.89)	(0.01)	(0.01)	(-2.35)	(-2.20)
Institutional Ownership	0.403***	-0.160***	0.069	0.265***	-0.170***	-0.016
	(10.89)	(-4.23)	(1.62)	(7.35)	(-5.04)	(-0.37)
Analyst Coverage	-0.173***	0.303***	0.136**	-0.177***	0.206***	0.173***
	(-2.98)	(5.10)	(1.99)	(-3.47)	(4.33)	(2.83)
Closing Price	-0.078	0.083	-0.028	0.143***	-0.100**	-0.070
	(-1.42)	(1.46)	(-0.43)	(2.97)	(-2.23)	(-1.26)
R& D Expenditure	-0.064	0.024	0.035	-0.065	0.007	0.049
	(-1.55)	(0.56)	(0.72)	(-1.64)	(0.19)	(1.03)
Total Assets	-0.157***	0.010	-0.017	-0.145***	0.035	-0.027
	(-3.91)	(0.25)	(-0.36)	(-2.99)	(0.77)	(-0.47)
Trading Volume	0.674***	-0.513***	-0.374***	0.614***	-0.362***	-0.312***
	(8.67)	(-6.45)	(-4.09)	(8.90)	(-5.61)	(-3.78)
Volatility	0.395***	0.404***	0.531***	0.392***	0.405***	0.500***
	(8.83)	(8.82)	(10.27)	(9.49)	(10.50)	(10.40)
Intangible assets	-0.255***	0.105**	0.073	-0.239***	0.034	0.113*
	(-4.99)	(2.00)	(1.24)	(-4.48)	(0.69)	(1.80)
Dividend	0.049	-0.009	0.008	0.050	-0.011	-0.031
	(1.30)	(-0.23)	(0.18)	(1.42)	(-0.34)	(-0.74)
Market Return	0.125***	-0.218***	-0.085**	0.060*	-0.229***	0.001
	(3.43)	(-5.82)	(-2.03)	(1.67)	(-6.78)	(0.03)
Adjusted R^2	0.536	0.514	0.470	0.498	0.562	0.420

Table 8
Alternative measure of internal governance: Industry adjusted age difference

This table reports the regression analysis of Stock market liquidity on internal governance after controlling for the inherent differences across industries. We adjust our measure of internal governance (Age difference) as follows: Industry adjusted age difference$_{i,t}$ = Age difference$_{i,t}$ – Age difference$_{ind,t}$, where, Age difference$_{i,t}$ is the difference between CEO's and top 4 subordinates ages for firm *i* and year *t*, Age difference$_{ind,t}$ is the average Age difference for the industry *i* in year *t*. We estimate the following model.

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1$ *Internal Governance (Industry Adjusted Age Difference)*$_{i,t} + \beta_2$ $Gov\text{-}Index_{i,t} + \beta_3$ $Institutional\ Ownership_{i,t} + \beta_4$ $Analyst\ Coverage_{i,t} + \beta_5$ $Log(Closing\ Price_{i,t}) + \beta_6$ $R\&D\ Expenditure_{i,t} + \beta_7$ $Total\ Assets_{i,t} + \beta_8$ $Log(Trading\ Volume_{i,t}) + \beta_9$ $Volatility_{i,t} + \beta_{10}$ $Intangible\ Asset_{i,t} + \beta_{11}$ $Dividend_{i,t} + \beta_{12}$ $Market\ Return_{i,t} + \varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (Industry Adjusted Age Difference)* is calculated as described above, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Turnover	Percentage Spread	Gibbs estimate
Internal Governance (Industry adjusted age difference)	0.023*	-0.034***	-0.026*
	(1.83)	(-2.83)	(-1.92)
Gov-Index	0.066***	-0.176***	-0.053***
	(4.78)	(-13.44)	(-3.61)
Institutional Ownership	0.265***	-0.134***	-0.009
	(19.80)	(-10.62)	(-0.63)
Analyst Coverage	-0.125***	0.187***	0.079***
	(-6.41)	(10.15)	(3.86)
Closing Price	0.091***	-0.087***	-0.069***
	(5.20)	(-5.29)	(-3.75)
R&D Expenditure	0.013	0.001	0.012
	(0.98)	(0.09)	(0.87)
Total Assets	-0.046***	0.011	0.031**
	(-3.22)	(0.79)	(2.00)
Trading Volume	0.451***	-0.284***	-0.222***
	(18.55)	(-12.38)	(-8.68)
Volatility	0.436***	0.375***	0.519***
	(28.81)	(26.27)	(32.56)
Intangible assets	-0.124***	0.045***	0.005
	(-8.05)	(3.08)	(0.30)
Dividend	-0.011	0.023*	-0.054***
	(-0.82)	(1.82)	(-3.77)
Market Return	0.082***	-0.237***	-0.037***
	(6.19)	(-18.92)	(-2.65)
Adjusted R^2	0.412	0.478	0.430

Table 9
Positive vs. Negative Industry-Adjusted Age Difference

To further investigate if industry-adjusted internal governance measure affects liquidity, we divide our sample firms into groups with positive and negative industry adjusted age differences and analyze the following regression model for each sub-sample:

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (age\ difference)_{i,t} + \beta_2\ Gov\text{-}Index_{i,t} + \beta_3\ Institutional\ Ownership_{i,t} + \beta_4\ Analyst\ Coverage_{i,t} + \beta_5\ Log(Closing\ Price_{i,t}) + \beta_6\ R\&D\ Expenditure_{i,t} + \beta_7\ Total\ Assets_{i,t} + \beta_8\ Log(Trading\ Volume_{i,t}) + \beta_9\ Volatility_{i,t} + \beta_{10}\ Intangible\ Asset_{i,t} + \beta_{11}\ Dividend_{i,t} + \beta_{12}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Panel A: Descriptive statistics: Internal Governance for the firms divided based on level of industry adjusted internal governance

Variable	N	Mean	Median	Minimum	Maximum	SD
Positive Industry adjusted measure (Firm-Industry)						
CEO Age	2239	58.01	57.50	40.00	91.00	5.59
Sub Managers' Age	2239	49.23	49.25	33.00	65.50	4.15
Internal Governance (Age difference)	2239	8.99	8.00	-3.50	36.33	4.77
Internal Governance (Industry adjusted age difference)	2239	5.25	4.20	0.00	31.97	4.42
Negative industry adjusted measure (Firm-Industry)						
CEO Age	2229	51.38	51.50	34.00	68.50	5.01
Sub Managers' Age	2229	52.07	52.00	39.33	70.00	4.20
Internal Governance (Age difference)	2229	-1.35	-0.50	-15.50	13.33	4.19
Internal Governance (Industry adjusted age difference)	2229	-5.11	-4.18	-17.90	0.00	3.95

Panel B. Summary of regression results						
Variable	Positive Age difference relative to the industry			Negative Age difference relative to the industry		
	Turnover	Percentage Spread	Gibbs estimate	Turnover	Percentage Spread	Gibbs estimate
Internal Governance (Industry adjusted age difference)	0.038**	-0.064***	-0.056***	-0.006	-0.036*	-0.009
	(2.13)	(-3.68)	(-2.86)	(-0.32)	(-1.69)	(-0.48)
Gov-Index	0.053***	-0.200***	-0.086***	0.082***	-0.147***	-0.015
	(2.76)	(-10.76)	(-4.05)	(4.13)	(-8.05)	(-0.77)
Institutional Ownership	0.250***	-0.124***	0.000	0.277***	-0.142***	-0.019
	(13.04)	(-6.75)	(0.02)	(14.64)	(-8.14)	(-1.02)
Analyst Coverage	-0.119***	0.140***	0.043	-0.126***	0.236***	0.110***
	(-4.35)	(5.37)	(1.45)	(-4.53)	(9.15)	(3.96)
Closing Price	0.083***	-0.091***	-0.074***	0.104***	-0.084***	-0.065***
	(3.31)	(-3.80)	(-2.73)	(4.26)	(-3.73)	(-2.63)
R&D Expenditure	-0.003	0.005	0.008	0.027	-0.005	0.018
	(-0.15)	(0.28)	(0.42)	(1.43)	(-0.26)	(0.93)
Total Assets	-0.076***	0.009	0.090***	-0.033	0.017	-0.012
	(-3.90)	(0.47)	(4.17)	(-1.59)	(0.89)	(-0.56)
Trading Volume	0.495***	-0.242***	-0.156***	0.417***	-0.337***	-0.300***
	(14.61)	(-7.44)	(-4.24)	(11.96)	(-10.44)	(-8.55)
Volatility	0.419***	0.376***	0.502***	0.441***	0.384***	0.538***
	(19.46)	(18.22)	(21.41)	(20.58)	(19.35)	(24.96)
Intangible assets	-0.144***	0.056***	-0.024	-0.109***	0.038*	0.041*
	(-6.96)	(2.80)	(-1.02)	(-4.85)	(1.83)	(1.67)
Dividend	-0.038*	0.052***	-0.052**	0.004	0.008	-0.055***
	(-1.94)	(2.82)	(-2.44)	(0.21)	(0.45)	(-2.87)
Market Return	0.085***	-0.255***	-0.036*	0.081***	-0.219***	-0.038**
	(4.57)	(-14.34)	(-1.79)	(4.26)	(-12.43)	(-1.97)
Adjusted R^2	0.433	0.479	0.402	0.401	0.487	0.472

Table 10

Alternative measure of internal governance: Options ratio

In this table we use stock options based measure, as an alternative measure of internal governance. We test the following regression model.

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (options\ ratio)_{i,t} + \beta_2\ Gov\text{-}Index_{i,t} + \beta_3\ Institutional\ Ownership_{i,t} + \beta_4\ Analyst\ Coverage_{i,t} + \beta_5\ Log(Closing\ Price_{i,t}) + \beta_6\ R\&D\ Expenditure_{i,t} + \beta_7\ Total\ Assets_{i,t} + \beta_8\ Log(Trading\ Volume_{i,t}) + \beta_9\ Volatility_{i,t} + \beta_{10}\ Intangible\ Asset_{i,t} + \beta_{11}\ Dividend_{i,t} + \beta_{12}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

Turnover is average daily volume as a proportion of shares outstanding, *Percentage Spread*, which is the percent quoted spread, *Gibbs estimate*, which is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (options ratio)* is the ratio of unexercised exercisable options of subordinate manager to unexercised exercisable options of the CEO, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, *Industry* is a dummy variable to control for industry fixed effects, *Year* is a dummy variable to control for time fixed effects, and $\varepsilon_{i,t}$ is the error term. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Turnover	Spread	Gibbs
Options Ratio	0.020***	-0.012**	-0.027***
	(3.10)	(-2.42)	(-4.67)
Gov-Index	0.045***	-0.149***	-0.040**
	(2.92)	(-10.37)	(-2.39)
Institutional Ownership	0.281***	-0.154***	-0.017
	(19.15)	(-10.00)	(-0.98)
Analyst Coverage	-0.132***	0.195***	0.072***
	(-5.67)	(10.00)	(3.49)
Closing Price	0.096***	-0.063***	-0.074***
	(4.5)	(-3.55)	(-3.97)
R&D Expenditure	0.011	0.000	0.013*
	(0.73)	(0.05)	(1.78)
Total Assets	-0.054***	0.010	0.042
	(-3.21)	(1.55)	(0.76)
Trading Volume	0.450***	-0.320***	-0.220***
	(15.53)	(-11.73)	(-7.15)
Volatility	0.444***	0.376***	0.512***
	(20)	(17.26)	(22.23)
Intangible assets	-0.122***	0.053***	0.001
	(-5.5)	(4.1)	(0.04)
Dividend	-0.020	0.024*	-0.048***
	(-1.53)	(1.89)	(-4.01)
Market Return	0.073***	-0.234***	-0.035**
	(5.29)	(-14.87)	(-2.03)
Adjusted R^2	0.4137	0.4668	0.4180

Table 11
Alternative measure of internal governance: CEO Pay Slice

This table reports the regression analysis of Stock market liquidity on internal governance using Bebchuk, Cremers and Peyer (2011) CEO pay slice as an alternative measure of internal governance. We estimate the following regression model:

Turnover$_{i,t}$ *or Percentage Spread*$_{i,t}$ *or Gibbs estimate*$_{i,t}$ = β_0 + β_1 *CEO Pay Slice*$_{i,t}$ + β_2 *Gov-Index*$_{i,t}$ + β_3 *Institutional Ownership*$_{i,t}$ + β_4 *Analyst Coverage*$_{i,t}$ +β_5 *Log(Closing Price*$_{i,t}$*)* + β_6 *R&D Expenditure*$_{i,t}$ + β_7 *Total Assets*$_{i,t}$ + β_8 *Log(Trading Volume*$_{i,t}$*)* + β_9 *Volatility*$_{i,t}$+ β_{10} *Intangible Asset*$_{i,t}$+ β_{11} *Dividend*$_{i,t}$ + β_{12} *Market Return*$_{i,t}$ + $\varepsilon_{i,t}$

where *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread* is the percent quoted spread, *Gibbs estimate* is a measure of effective cost calculated using Gibb's sampler, *CEO Pay Slice* is the ratio of CEO pay to the total pay of top 5 managers and is an inverse measure of internal governance, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *Closing Price* is the mean daily stock price at the close, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, and $\varepsilon_{i,t}$ is the error term. The subscripts *i* and *t* refers to stock *i* and year *t*. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

	Turnover	Percentage Spread	Gibbs
Internal Governance	-0.025*	0.044***	0.021
(CEO Pay Slice)	(-1.86)	(3.52)	(1.50)
Gov-Index	0.064***	-0.164***	-0.047***
	(4.45)	(-12.19)	(-3.15)
Institutional Ownership	0.281***	-0.131***	-0.018
	(20.44)	(-10.16)	(-1.27)
Analyst Coverage	-0.148***	0.188***	0.078***
	(-7.29)	(9.92)	(3.71)
Closing Price	0.098***	-0.086***	-0.071***
	(5.48)	(-5.13)	(-3.80)
R& D Expenditure	0.010	0.002	0.012
	(0.76)	(0.19)	(0.86)
Total Assets	-0.045***	0.012	0.034**
	(-3.03)	(0.87)	(2.11)
Trading Volume	0.462***	-0.278***	-0.226***
	(18.41)	(-11.88)	(-8.64)
Volatility	0.442***	0.392***	0.519***
	(27.95)	(26.54)	(31.47)
Intangible assets	-0.132***	0.050***	0.004
	(-8.27)	(3.37)	(0.21)
Dividend	-0.016	0.016	-0.048***
	(-1.14)	(1.23)	(-3.29)
Market Return	0.080***	-0.234***	-0.033**
	(5.76)	(-18.08)	(-2.25)
Adjusted R^2	0.409	0.484	0.435

Table 12

Internal governance and stock market liquidity: Regression results using changes in variables

This table presents the results using changes in variables. We examine whether yare to year changes in liquidity can be explained by changes in internal governance.

Δ *Turnover*$_{i,t}$ *or* Δ *Percentage Spread*$_{i,t}$ *or* Δ *Gibbs estimate*$_{i,t}$ = β_0 + β_1 Δ *Internal Governance (age difference)*$_{i,t}$ + β_2 *Gov-Index*$_{i,t}$ + β_3 *Institutional Ownership*$_{i,t}$ + β_4 *Analyst Coverage*$_{i,t}$ +β_5 *Log(Closing Price*$_{i,t}$*)* + β_6 *R&D Expenditure*$_{i,t}$ + β_7 *Total Assets*$_{i,t}$ + β_8 *Log(Trading Volume*$_{i,t}$*)* + β_9 *Volatility*$_{i,t}$+ β_{10} *Intangible Asset*$_{i,t}$+ β_{11} *Dividend*$_{i,t}$ + β_{12} *Market Return*$_{i,t}$ + $\varepsilon_{i,t}$

Δ denotes change in the variable *Turnover* is average daily volume as a proportion of shares outstanding, *Percentage Spread*, which is the percent quoted spread, *Gibbs estimate*, which is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, and *Market Return* is the daily return on S&P index. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Δ Turnover	Δ Percentage Spread	Δ Gibbs estimate
Δ Internal Governance (Age difference)	-0.015	-0.028*	-0.040**
	(-0.86)	(-1.71)	(-2.06)
Gov-Index	0.102***	0.298***	0.101***
	(5.45)	(16.7)	(4.73)
Institutional Ownership	0.038**	0.085***	-0.002
	(2.09)	(4.97)	(-0.08)
Analyst Coverage	-0.252***	-0.072***	0.05^{8*}
	(-9.44)	(-2.82)	(1.92)
Closing Price	0.047**	0.219***	0.155***
	(2.01)	(9.88)	(5.85)
R&D Expenditure	-0.029	-0.023	0.007
	(-1.63)	(-1.39)	(0.35)
Total Assets	0.018	-0.006	-0.050**
	(0.9)	(-0.31)	(-2.16)
Trading Volume	0.272***	0.038	-0.136***
	(8.22)	(1.2)	(-3.64)
Volatility	0.312***	-0.111***	0.117***
	(15.21)	(-5.71)	(5.05)
Intangible assets	-0.050**	0.003	0.065**
	(-2.37)	(0.16)	(2.53)
Dividend	0.085***	-0.005	0.039*
	(4.64)	(-0.27)	(1.85)
Market Return	0.007	-0.164***	-0.133***
	(0.4)	(-9.56)	(-6.54)
Adjusted R^2	0.1371	0.2207	0.0485

Table 13

Internal governance and stock market liquidity: Age difference increase

In this table, use the increase in age difference as an exogenous shock to test the relationship between internal governance and stock market liquidity. We analyze the following regression.

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (options\ ratio)_{i,t} + \beta_2\ Age\ difference\ increase_{i,t} + \beta_3\ Gov\text{-}Index_{i,t} + \beta_4\ Institutional\ Ownership_{i,t} + \beta_5\ Analyst\ Coverage_{i,t} + \beta_6\ Log(Closing\ Price_{i,t}) + \beta_7\ R\&D\ Expenditure_{i,t} + \beta_8\ Total\ Assets_{i,t} + \beta_9\ Log(Trading\ Volume_{i,t}) + \beta_{10}\ Volatility_{i,t} + \beta_{11}\ Intangible\ Asset_{i,t} + \beta_{12}\ Dividend_{i,t} + \beta_{13}\ Market\ Return_{i,t} + \varepsilon_{i,t}$

Turnover is average daily volume as a proportion of shares outstanding, *Percentage Spread*, which is the percent quoted spread, *Gibbs estimate*, which is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Age difference increase*$_{i,t}$ is a dummy variable equals 1 only if there is an increase in age difference between the CEO and subordinate managers from year to year.*Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, *Industry* is a dummy variable to control for industry fixed effects, *Year* is a dummy variable to control for time fixed effects, and $\varepsilon_{i,t}$ is the error term. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Turnover	Spread	Gibbs
Internal Governance	0.035***	-0.034**	-0.027*
(Age difference)	(2.85)	(-2.36)	(-1.90)
Age difference increase	0.018	-0.042***	-0.020*
	(1.38)	(-4.58)	(-1.66)
Gov-Index	0.066***	-0.174***	-0.052***
	(4.64)	(-12.8)	(-3.51)
Institutional Ownership	0.265***	-0.133***	-0.008
	(19.83)	(-9.72)	(-0.59)
Analyst Coverage	-0.127***	0.188***	0.080***
	(-5.61)	(10.49)	(4.22)
Closing Price	0.091***	-0.088***	-0.069***
	(4.21)	(-5.01)	(-3.95)
R&D Expenditure	0.013	0.001	0.012**
	(0.92)	(0.07)	(2.12)
Total Assets	-0.047**	0.012**	0.032
	(-2.5)	(1.99)	(0.72)
Trading Volume	0.452***	-0.283***	-0.221***
	(16.3)	(-10.94)	(-7.7)
Volatility	0.435***	0.375***	0.519***
	(20.25)	(18.53)	(22.58)
Intangible assets	-0.123***	0.044***	0.004
	(-6.13)	(4.03)	(0.17)
Dividend	-0.011	0.024**	-0.053***
	(-0.88)	(2.14)	(-4.56)
Market Return	0.081***	-0.234***	-0.035**
	(6.47)	(-16.76)	(-2.19)
Adjusted R^2	0.4110	0.4785	0.4281

Table 14

Internal governance and stock market liquidity: adjustment for industry and time fixed effects

To control for differences across industries and across time on the liquidity and level of monitoring, we analyze the following regression model:

$Turnover_{i,t}$ *or* $Percentage\ Spread_{i,t}$ *or* $Gibbs\ estimate_{i,t} = \beta_0 + \beta_1\ Internal\ Governance\ (age\ difference)_{i,t} + \beta_2\ Gov\text{-}Index_{i,t} + \beta_3\ Institutional\ Ownership_{i,t} + \beta_4\ Analyst\ Coverage_{i,t} + \beta_5\ Log(Closing\ Price_{i,t}) + \beta_6\ R\&D\ Expenditure_{i,t} + \beta_7\ Total\ Assets_{i,t} + \beta_8\ Log(Trading\ Volume_{i,t}) + \beta_9\ Volatility_{i,t} + \beta_{10}\ Intangible\ Asset_{i,t} + \beta_{11}\ Dividend_{i,t} + \beta_{12}\ Market\ Return_{i,t} + \sum_{i=1}^{48} \delta_i Industry_i + \sum_{j=1}^{7} \theta_j Year_j + \varepsilon_{i,t}$

Turnover is average daily volume as a proportion of shares outstanding, *Percentage Spread*, which is the percent quoted spread, *Gibbs estimate*, which is a measure of effective cost calculated using Gibb's sampler, *Internal Governance (age difference)* is the difference between *CEO Age* and *Subordinate Manager Age*, *Gov-Index* is Chung, Elder and Kim (2010) governance index, *Institutional Ownership* is the proportion of outstanding stocks held by institutional investors, *Analyst Coverage* is the mean number of analysts following a firm, *R&D Expenditure* is a firm's annual expenditure on research and development activities, *Total Assets* is the book value of a firm's total assets, *Trading Volume* is the mean daily dollar trading volume, *Volatility* is the standard deviation of daily returns, *Intangible Assets* is the book value of total intangible assets, *Dividend* is the annual dividend paid per share, *Market Return* is the daily return on S&P index, *Industry* is a dummy variable to control for industry fixed effects, and $\varepsilon_{i,t}$ is the error term. T-statistics calculated using White's corrected standard errors are reported in parentheses. Levels of significance are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Variable	Turnover	Percentage Spread	Gibbs estimate
Internal Governance	0.015**	-0.032***	-0.017**
(Age difference)	(1.98)	(-2.87)	(-2.11)
Gov-Index	0.422***	-1.542***	-0.239***
	(5.59)	(-14.14)	(-3.07)
Institutional Ownership	0.670***	-0.449***	-0.008
	(9.43)	(-9.01)	(-0.22)
Analyst Coverage	-0.168***	0.330***	0.098***
	(-7.80)	(10.64)	(4.35)
Closing Price	0.264***	-0.385***	-0.217***
	(4.43)	(-4.47)	(-3.52)
R&D Expenditure	-0.001	-0.001	0.003
	(-0.08)	(-0.10)	(0.39)
Total Assets	-0.023***	0.009	0.027***
	(-2.78)	(0.76)	(3.08)
Trading Volume	2.765***	-2.670***	-1.462***
	(9.08)	(-12.76)	(-9.63)
Volatility	0.584***	0.932***	0.719***
	(25.32)	(27.99)	(30.28)
Intangible assets	-0.055***	0.040***	0.006
	(-5.92)	(2.97)	(0.60)
Dividend	-0.018*	0.012	-0.024**
	(-1.95)	(0.93)	(-2.43)
Market Return	0.041***	-0.189***	-0.017**
	(5.73)	(-18.48)	(-2.43)
Industry Fixed Effect	Yes	Yes	Yes
Adjusted R^2	0.838	0.663	0.849

Appendix A
Variable Description and source

Variable	Description	Source
CEO Age	Mean CEO age for each firm each year	EXECUCOMP
Subordinate Manager Age	Mean age for top 4 managers for each firm each year	EXECUCOMP
Relative Age Difference	Difference between the mean CEO age and mean subordinate manager age	EXECUCOMP
CEO Pay Slice	Ratio of CEO total compensation to the top 5 executives' total compensation.	EXECUCOMP
Option Ratio	Average subordinates' percentage of exerciable unexercised option to the CEO percentage.	EXECUCOMP
Gov-Index	Chung, Elder and Kim (2010) Governance Index	Institutional Shareholders Services (ISS)
Institutional Ownership (%)	Proportions of stocks held by institutions	Spectrum CDA
Analyst coverage	Number of analysts following a firm	I/B/E/S
Gibbs estimate (%)	Hasbrouck's Gibbs estimate using Gibbs Sampler	Hasbrouck's website
Percent Spread	Quoted spread as a proportion of quote midpoint	CRSP
Amihud Price Impact	Daily Absolute Return divided by turnover averaged over year	CRSP
Turnover Ratio	Dollar volume per share outstanding	CRSP
S&P index return (%)	Daily return on S and P 500 index	CRSP
Closing Price	Daily closing price for a stock	CRSP
R& D Expenditure(Millions)	Annual firm expense on research and development activity	COMPUSTAT
Dollar Volume (billions)	Daily dollar volume	CRSP
Total Assets (millions)	Firm's total assets	COMPUSTAT
Volatility (%)	Standard deviation of returns	CRSP
Intangible assets (Millions)	Total intangible assets	COMPUSTAT
Dividend per share	Dollar dividend paid per share outstanding	COMPUSTAT